

Major Project-A Report
On
**Adaptive Continuous User Authentication Using
Behavioral Biometrics**

Submitted in partial fulfillment of the requirement of
University of Mumbai for the Degree of

Bachelor of Technology
In
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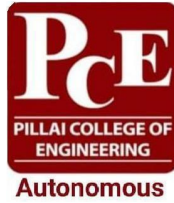
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We declare that this written submission for B.Tech. Declaration entitled “**Adaptive Continuous User Authentication Using Behavioral Biometrics**” represents our ideas in our own words and where others' ideas or words have been included. We have adequately cited and referenced the original sources. We also declared that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any ideas / data / fact / source in our submission. We understand that any violation of the above will cause disciplinary action by the institute and also evoke penal action from the sources which have thus not been properly cited or from whom paper permission has not been taken when needed.

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Abstract

With the rapid growth of online services and digital platforms, secure user authentication has become a critical requirement in cybersecurity. Traditional authentication mechanisms such as passwords, PINs, and one-time verification methods are vulnerable to attacks and fail to verify the user continuously after login. Behavioral biometric-based authentication has emerged as an effective solution for continuous verification by analyzing unique user interaction patterns such as typing behavior and mouse movements. Various techniques including machine learning and deep learning approaches such as Support Vector Machines (SVM), Random Forest, and Neural Networks have been explored for behavioral authentication. However, existing systems suffer from limitations such as behavioral drift, high false rejection rates, inability to adapt to context changes, and vulnerability to profile poisoning attacks. To address these challenges, this project proposes an Adaptive Continuous User Authentication System using Behavioral Biometrics that continuously verifies the identity of users based on keyboard and mouse interaction patterns.

The system extracts behavioral features such as keystroke timing, flight time, typing rhythm, cursor speed, and trajectory patterns. Feature selection using Minimum Redundancy Maximum Relevance (mRMR) is applied to select the most relevant features for authentication. The system employs LightGBM and One-Class SVM models to distinguish genuine users from impostors while incorporating adaptive thresholds and hybrid user profiles to handle behavior variations. The proposed approach aims to improve authentication accuracy, reduce false rejection rates, and provide a reliable continuous authentication mechanism for secure applications.

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Chapter 1

INTRODUCTION

1.1 Fundamentals

With the rapid growth of digital platforms and online services, secure user authentication has become a critical requirement in modern cybersecurity systems. Traditional authentication techniques such as passwords, personal identification numbers (PINs), and one-time passwords (OTP) are widely used to verify user identity. However, these methods verify users only during the login phase and fail to ensure that the same user continues to operate the system throughout the session. This limitation creates security risks such as session hijacking, account takeover attacks, and unauthorized access after login.

To overcome these limitations, continuous authentication systems have been introduced. Continuous authentication verifies the identity of a user throughout an active session instead of relying only on a single login event. Behavioral biometrics has emerged as an effective technique for implementing continuous authentication because it analyzes unique patterns in human-computer interaction.

Behavioral biometrics refers to the identification of individuals based on behavioral patterns such as typing rhythm, mouse movement, touchscreen gestures, voice patterns, and interaction behavior with devices. Among these, keystroke dynamics and mouse dynamics are widely used because they can be collected passively without requiring additional hardware. Each user interacts with input devices in a unique manner, creating identifiable behavioral signatures.

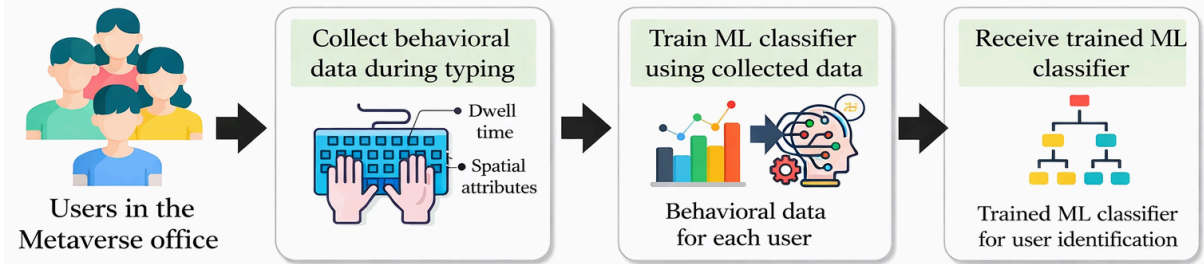
Several machine learning and deep learning techniques have been explored for behavioral biometric authentication. Commonly used methods include Support Vector Machines (SVM), Random Forest, Neural Networks, and anomaly detection models. These techniques analyze behavioral features such as keystroke timing, typing speed, cursor movement speed, trajectory patterns, and click behavior to distinguish genuine users from impostors.

However, existing behavioral biometric systems still face challenges such as behavioral drift, context variations, and high false rejection rates. Human behaviour naturally changes due to factors such as fatigue, stress, device changes, and environmental conditions. These variations make it difficult for traditional authentication models to maintain high accuracy over long periods of time.

To address these challenges, this project proposes an Adaptive Continuous User Authentication System using Behavioral Biometrics. The system continuously monitors keyboard and mouse interaction patterns from a secure web application, extracts behavioral features, and uses

machine learning models to verify the identity of the user throughout the session. By incorporating adaptive mechanisms and behavioral analysis, the system aims to improve security while reducing false authentication errors.

1. Creating authentication profile



2. Authenticating the user

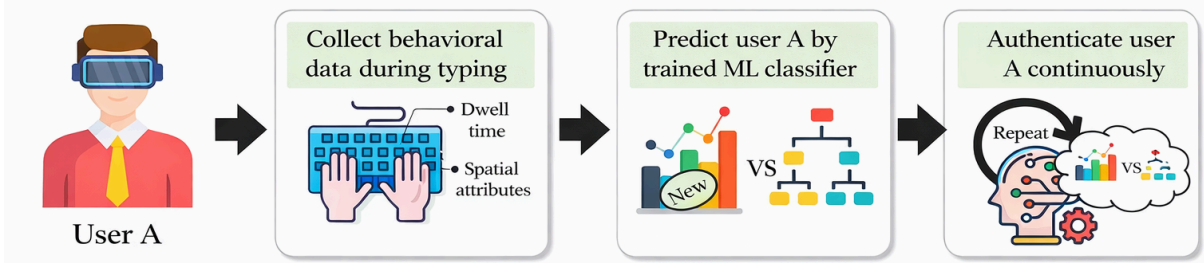


Fig 1.1: Continuous Authentication using Behavioral Biometrics

Figure 1.1 illustrates the concept of continuous authentication using behavioral biometrics. The system continuously collects keyboard and mouse interaction data from the user and analyzes behavioral features such as typing rhythm and mouse movement patterns. These features are processed by machine learning models to verify the user's identity during the entire session. If abnormal behaviour is detected, the system can trigger additional authentication or terminate the session.

1.2 Objectives

The objective of this work is to design and implement an adaptive continuous authentication system using behavioral biometrics to enhance user security in digital systems. The specific objectives of this project are as follows:

1. To study existing authentication techniques and behavioral biometric approaches used for continuous user authentication.
2. To analyze the limitations of traditional authentication systems such as passwords and one-time verification methods.
3. To design a behavioral biometric authentication system using keyboard dynamics and mouse movement patterns.
4. To extract relevant behavioral features such as keystroke timing, typing rhythm, cursor speed, and movement trajectories.

5. To apply machine learning techniques such as LightGBM and One-Class SVM for user authentication.
6. To implement feature selection techniques such as Minimum Redundancy Maximum Relevance (mRMR) for selecting important behavioral features.
7. To develop an adaptive authentication framework that can handle behaviour variations and reduce false rejection rates.

1.3 Scope

The scope of this project is to design and implement an adaptive continuous authentication system based on behavioral biometrics. The system focuses on analyzing keyboard typing patterns and mouse movement behaviour to verify user identity during an active session. The proposed system will be implemented through a web-based application that collects behavioral interaction data from users while they interact with the system.

The system aims to provide enhanced security by continuously verifying user identity and detecting potential unauthorized access during the session. The project includes the design of feature extraction methods, implementation of machine learning models, and evaluation of authentication performance using appropriate metrics. The intended users of this system include individuals accessing secure web applications such as online banking systems, enterprise platforms, and other digital services that require continuous authentication.

1.4 Outline

The report is organized into several chapters to present the work carried out in this project. Chapter 1 provides an introduction to the project and explains the fundamental concepts related to behavioral biometric authentication. It also presents the objectives and scope of the project. Chapter 2 presents a review of existing literature related to continuous authentication systems and behavioral biometric techniques, along with their advantages and limitations. Chapter 3 describes the theoretical background and the proposed system architecture for the adaptive continuous authentication framework. Chapter 4 discusses the implementation details, experimental setup, and applications of the proposed system. Finally, Chapter 5 summarizes the findings of the project and presents the conclusion along with possible directions for future work.

Chapter 2

Literature Survey

In this chapter, the relevant techniques and research works related to behavioral biometric authentication are reviewed. It describes various approaches used for continuous authentication using behavioral biometrics such as keystroke dynamics and mouse movement analysis. The current literature related to behavioral biometric authentication systems is examined to identify different techniques developed in this domain. The advantages and limitations of these techniques are also discussed. The purpose of this review is to understand the existing approaches, identify research gaps, and motivate the development of improved authentication systems.

A literature review is an objective and critical summary of previously published research relevant to the topic under consideration. Its purpose is to create familiarity with current research trends and methods in behavioral biometric authentication and to justify the need for improved adaptive authentication systems.

A typical literature review consists of an introduction, detailed analysis of techniques, discussion of advantages and limitations, and a summary of findings.

2.1 Keystroke Dynamics Based Authentication Techniques

Keystroke dynamics is one of the most widely studied behavioral biometric techniques used for continuous authentication. This technique analyzes typing behavior patterns such as key press duration, typing speed, and time intervals between keystrokes. Each individual has a unique typing rhythm which can be used to verify identity.

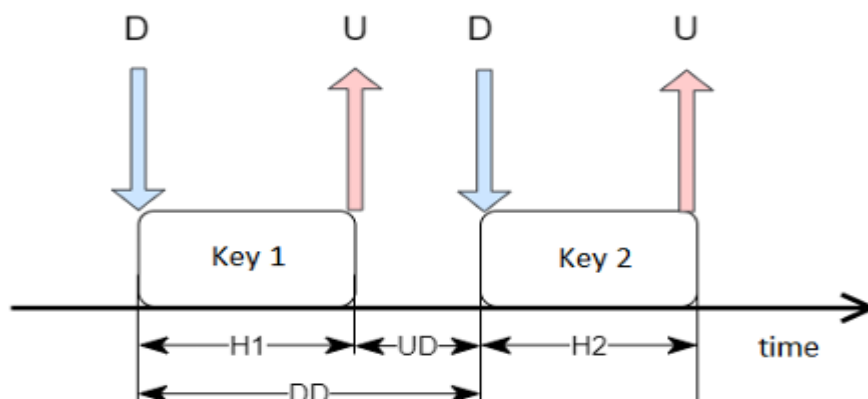


Fig 2.1: Support Vector Machine Based Authentication

Support Vector Machine (SVM) is one of the most commonly used machine learning techniques for behavioral biometric authentication. Kasprowski et al. [2] used SVM models for keystroke dynamics-based user authentication. The system analyzes keystroke timing features such as dwell time and flight time to distinguish between genuine users and impostors.

This technique represents the state-of-the-art approach for behavioral authentication systems and provides effective classification between user behaviour patterns. The method extracts keystroke features and trains machine learning models to identify unique behavioral signatures.

Advantages:

- High classification accuracy for behavioral patterns
- Works well with small and medium-sized datasets
- Effective in distinguishing genuine users from impostors

Disadvantages:

- Performance decreases when user behaviour changes over time
- Sensitive to noise and variations in typing patterns
- Requires feature engineering and preprocessing

2.1.2 Neural Network Based Authentication

Neural network models have also been used for behavioral biometric authentication. Baig et al. [3] proposed a neural network-based authentication system that learns user behaviour patterns from keystroke data. Deep learning techniques such as Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN) have been applied to capture complex temporal patterns in typing behaviour.

Neural networks are capable of learning complex relationships between behavioral features and user identity, making them useful for large datasets.

Advantages:

- Ability to learn complex behavioral patterns
- High performance with large datasets
- Automatic feature learning capability

Disadvantages:

- Requires large training datasets
- High computational cost
- Difficult to interpret model decisions

2.1.3 LightGBM-based authentication

achieves superior classification accuracy and speed through efficient histogram-based learning, but remains highly sensitive to performance degradation caused by user typing fluctuations from fatigue, stress, or physical injuries. Without continuous monitoring and specialized

adaptation, these models struggle to remain reliable across diverse keyboard layouts and linguistic variations.

2.1.4 One-class SVM-based authentication

excels at detecting anomalies when labelled impostor data is unavailable, yet it often prioritizes identifying outliers at the expense of accurately recognizing the legitimate user's own behavioral variations. These unsupervised models require intensive hyperparameter tuning (such as optimizing the RBF kernel) and frequently result in higher false negative rates because they struggle to model the inherent inconsistency of human patterns over long periods

2.2 Mouse Dynamics Based Authentication Techniques

Mouse dynamics is another behavioral biometric technique used for continuous authentication. It analyzes how users interact with the mouse while navigating the system. Features such as cursor movement speed, trajectory patterns, click frequency, and movement acceleration are used to identify users.

2.2.1 Mouse Movement Pattern Analysis

Wang et al. [4] applied mouse movement analysis for continuous authentication. The system captures cursor movement patterns and extracts behavioral features such as velocity, acceleration, and direction changes. These features are used to build authentication models that distinguish between genuine users and impostors.

The system introduces a behavioral modeling process that includes data collection, feature extraction, and classification.

Advantages:

- Non-intrusive data collection
- Can operate continuously during user sessions
- Does not require additional hardware

Disadvantages:

- Behavior may vary due to different devices or environments
- Requires continuous data collection
- Accuracy may decrease with limited interaction data

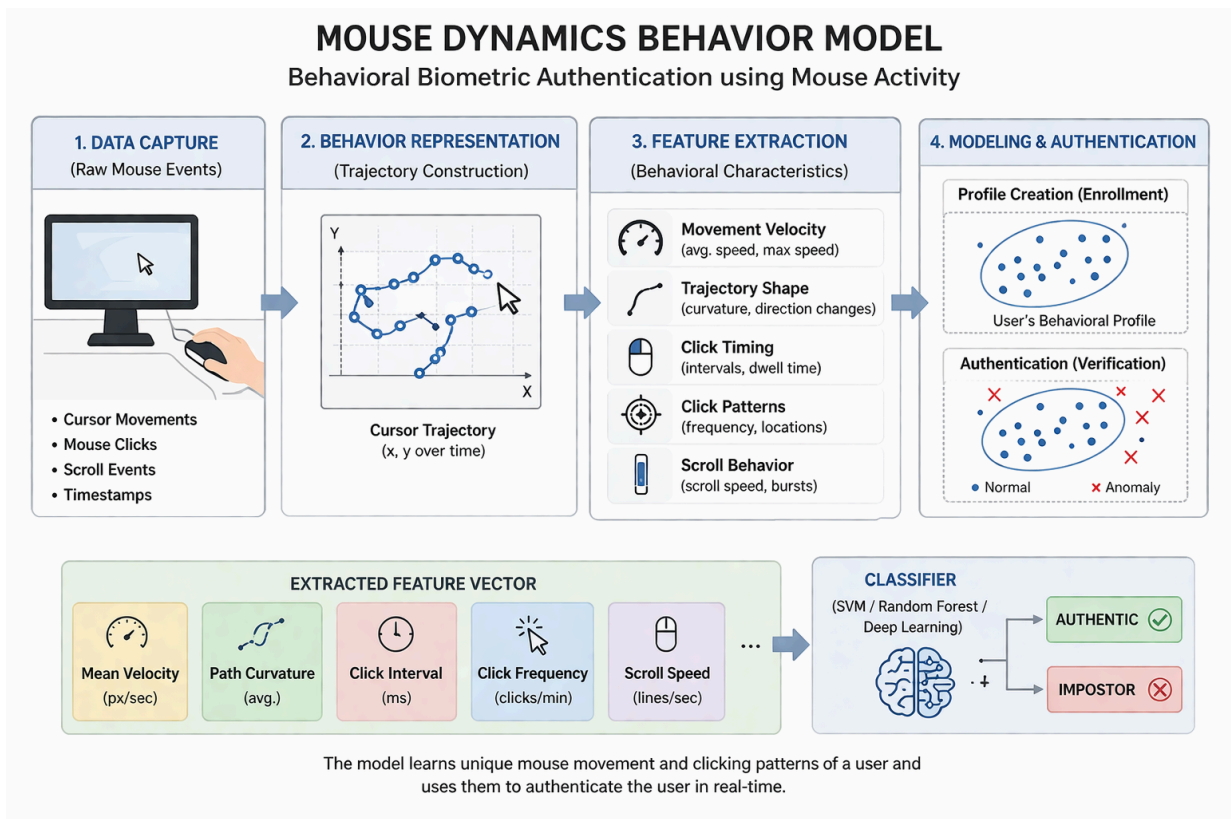


Fig 2.2: Mouse Behavioral Authentication

Hybrid authentication techniques combine multiple behavioral biometric signals such as keystroke dynamics and mouse movement patterns to improve authentication accuracy. Author Four et al. [5] proposed a hybrid authentication system that integrates keyboard and mouse behavioral features.

The architecture of the hybrid system is illustrated in Figure 2.1, where behavioral data is collected from user interactions, processed for feature extraction, and analyzed using machine learning models.

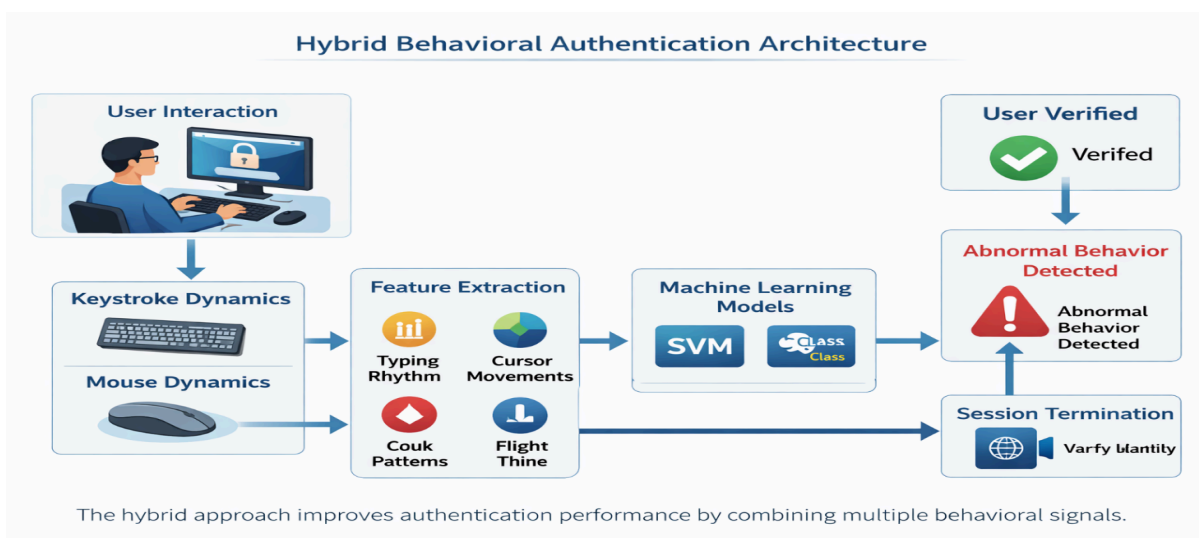


Fig 2.3: Hybrid Behavioral Authentication Architecture

The hybrid approach improves authentication performance by combining multiple behavioral signals and reducing the weaknesses of individual techniques.

Advantages:

- Improved authentication accuracy
- Combines strengths of multiple behavioral signals
- Reduces single-modality limitations

Disadvantages:

- Increased computational complexity
- Requires integration of multiple data sources
- More complex feature extraction process

2.3 Literature Summary

The literature survey highlights the various behavioral biometric techniques used for continuous user authentication. Several approaches including keystroke dynamics, mouse movement analysis, and hybrid authentication systems have been proposed to improve security in digital systems. Machine learning models such as Support Vector Machines and neural networks are widely used for behavioral pattern recognition.

However, existing systems still face several limitations such as behavioral drift, context variations, high false rejection rates, and lack of adaptability to changing user behaviour. These limitations motivate the development of adaptive authentication systems that can handle behavioral changes while maintaining high authentication accuracy.

Table 2.1 Summary of literature survey

Sr No	Paper Details & Reference	Problem Addressed	Methodology Used	Advantages	Limitations
1	Z. Jorgensen and T. Yu, "On Mouse Dynamics as a Behavioral Biometric for Authentication," ASIA CCS '11, pp. 475–482, 2011.	Mouse behavior authentication results often over-optimistic	Controlled experiments comparing mice vs touchpads.	Demonstrate detectable behavioral patterns.	Highly sensitive to hardware differences.

2	S. Krishnamoorthy et al., "Identification of User Behavioral Biometrics for Authentication Using Keystroke Dynamics and Machine Learning," ICBEA '18, pp. 50–57, 2018.	Identifying important biometric features for classification.	Used mRMR feature selection with SVM (RBF kernel).	Found touch pressure and size highly influential.	Limited to static keystroke dynamics.
3	X. Wang et al., "User Authentication Method Based on Keystroke Dynamics and Mouse Dynamics with Scene-Irrelated Features in Hybrid Scenes," Sensors, vol. 22, no. 17, art. no. 6627, Sept. 2022.	Accuracy drops when users perform different tasks.	Proposed SIURUA algorithm using Multiple Kernel Learning.	Reduced scene impact with 84% accuracy.	Requires very high feature dimension and long observation time.

4	P. Kasproski, Z. Borowska, and K. Harezlak, "Biometric Identification Based on Keystroke Dynamics," Sensors, vol. 22, no. 9, art. no. 3158, Apr. 2022.	Unclear impact of neural network architecture on identification accuracy.	Tested CNN, LSTM, and GRU combinations.	Mixed CNN-RNN architectures gave best results.	Accuracy decreases with more users.
5	A. F. Baig, S. Eskeland, and B. Yang, "Privacy-Preserving Continuous Authentication Using Behavioral Biometrics," International Journal of Information Security, vol. 22, pp. 1833–1847, 2023.	Authentication servers collect sensitive biometric data.	Used homomorphic encryption and oblivious transfer protocols.	Protects biometric data privacy.	Higher execution time due to cryptographic overhead.

6	U. Uslu, O. D. Incel, and G. I. Alptekin, "Evaluation of Deep Learning Models for Continuous Authentication Using Behavioral Biometrics," <i>Procedia Computer Science</i> , vol. 225, pp. 1272–1281, 2023.	Intruders can access sessions if users forget to log out.	Tested MLP, LSTM, and Conv-LSTM models on motion and touch data.	MLP achieved 99.85% accuracy.	Deep learning models consume high device resources.
7	X. Wang and D. Hou, "Enhancing Keystroke Dynamics Authentication with Ensemble Learning and Data Resampling Techniques," <i>Electronics</i> , vol. 13, no. 22, art. no. 4559, Nov. 2024.	Keystroke dynamics often suffer from data imbalance where legitimate samples far exceed impostor samples.	Combined SMOTE resampling with ensemble methods (Random Forest, XGBoost, Bagging) using a Django-based collection platform.	Improved classification of minority class (impostors), strengthening system security.	SMOTE has quadratic computational complexity $O(n^2)$ making it resource intensive.

8	L. Wyciślik, P. Wyleżek, and A. Momot, "Improved Biometric Identification of Keystroke Dynamics Based on Deep Learning Approaches," Sensors, vol. 24, no. 12, art. no. 3763, June 2024.	Lack of benchmark datasets for fair algorithm comparison.	CNN + GRU classifier using Buffalo dataset.	Achieved False Positive Rate of 1.91%.	Higher False Negative Rate due to pattern detection limits.
9	P. Bansal and A. Ouda, "Continuous Authentication in the Digital Age: An Analysis of Reinforcement Learning and Behavioral Biometrics," Computers, vol. 13, no. 4, art. no. 103, Apr. 2024.	Static authentication methods fail against modern cyberattacks.	Reinforcement Learning framework using Double Deep Q-Networks.	System adapts to behavior changes dynamically.	Accuracy dropped from training to testing due to inconsistent typing patterns.

10	L. Yang and S. Qin, "Identify User Features Impacting Keystroke, Mouse and Touchscreen Dynamics," Multimedia Tools and Applications, vol. 84, pp. 48685–48713, 2025.	Unclear which user characteristics influence behavioral biometrics.	Questionnaire survey of 107 participants.	Proposed personalized emotion recognition framework.	Relies on subjective self-reported data.
11	A. Martins et al., "Keystroke Dynamics for Intelligent Biometric Authentication with Machine Learning," Discover Applied Sciences, vol. 7, art. no. 992, 2025.	Increasing data breaches require stronger authentication layers.	Benchmarked KeyRecs dataset using KNN, RF, and LGBM models.	LGBM achieved 80% F1-score and lowest Equal Error Rate.	Datasets may not reflect environmental noise in real usage.

12	A. Muralidharan, A. Eaman, and E. Hassan, "A Comparative Analysis of Machine Learning Models for Behavioral Biometric Authentication using Keystroke Dynamics," <i>Procedia Computer Science</i> , vol. 265, pp. 116–123, 2025.	Traditional passwords are vulnerable to brute force; need for non-intrusive alternatives.	Evaluated SVM, Random Forest, XGBoost, LightGBM, CNN, and LSTM using CMU dataset.	LightGBM achieved highest accuracy (94.68%).	CMU dataset uses single-word password which may not reflect real-world typing.
13	K. Asgarov, "Endpoint Behaviour Modelling from User Interaction Metrics via Unsupervised Deep Learning for Anomaly Detection," <i>Azerbaijan Technical University</i> , 2025.	Active sessions are vulnerable to post-login takeovers.	Used unsupervised DL models (Autoencoders, Deep SVDD, DAGMM).	Multimodal autoencoders achieved F1-score of 0.92.	Study limited to 12 participants.

14	A. Almeahmadi, "Cyber Coercion Detection Using LLM-Assisted Multimodal Biometric System," Applied Sciences, vol. 15, no. 19, art. no. 10658, Oct. 2025.	Hard to detect coerced user actions when legitimate credentials are used.	Combined LLM reasoning with multimodal biometrics (keystroke, mouse, heart rate).	Achieved near-perfect detection (ROC-AUC = 1.00).	Tested mainly on simulated datasets.
15	J. Zhang et al., "SoundAuth: Non-Intrusive User Authentication via Keystroke Acoustic Biometrics with Triplet Networks," UESTC, 2025.	Software monitoring methods are vulnerable to keylogging.	Uses keystroke acoustic signatures via microphones.	Achieved 98% recognition rate.	Feature degradation when adding new users.

2.4 Research Gap

- Continuous authentication raises significant privacy concerns by potentially exposing sensitive user activity data
- High-performance deep learning models often require excessive computational power that is impractical for real-time edge devices
- The impact of emotions and task urgency on behavioral dynamics remains largely under-investigated
- Many authentication approaches require impractical collection times before a reliable decision can be reached
- Systems often experience accuracy drops when a user switches hardware, such as from a mouse to a touchpad
- Small participant sample sizes in most studies limit the generalizability of findings to

diverse populations

- Static models fail to adapt to long-term behavioral changes caused by natural factors like aging or skill progression
- Many systems are limited to fixed-text passwords rather than reflecting the complexity of real-world free-text activities
- A lack of universally accepted benchmark datasets prevents researchers from reliably comparing different identification algorithms
- Low-level biometrics struggle to distinguish between genuine security threats and routine workplace stress
- Research often fails to control for environmental variables such as screen resolution or specific device hardware
- Models are vulnerable to intentional imitation where an impostor mimics the legitimate user's specific tasks

Chapter 3

Proposed System

3.1 Overview

This chapter presents the proposed system architecture for the Adaptive Continuous User Authentication System using Behavioral Biometrics. The system continuously verifies the identity of users by analysing their interaction patterns with input devices such as keyboards and mouse. Behavioral biometrics provides a non-intrusive authentication mechanism that operates during an active session without interrupting the user.

Traditional authentication methods such as passwords and PINs provide only single-point authentication at login. Once access is granted, these methods cannot detect whether the current user is still the legitimate user. Continuous authentication systems address this limitation by monitoring behavioral patterns throughout the user session.

Behavioral biometrics includes several techniques such as keystroke dynamics, mouse dynamics, touchscreen gestures, and sensor-based biometrics. Among these techniques, keystroke dynamics and mouse movement patterns are widely used because they can be collected using existing hardware without requiring additional devices.

The classification of behavioral biometric authentication techniques is shown in Figure 3.1.

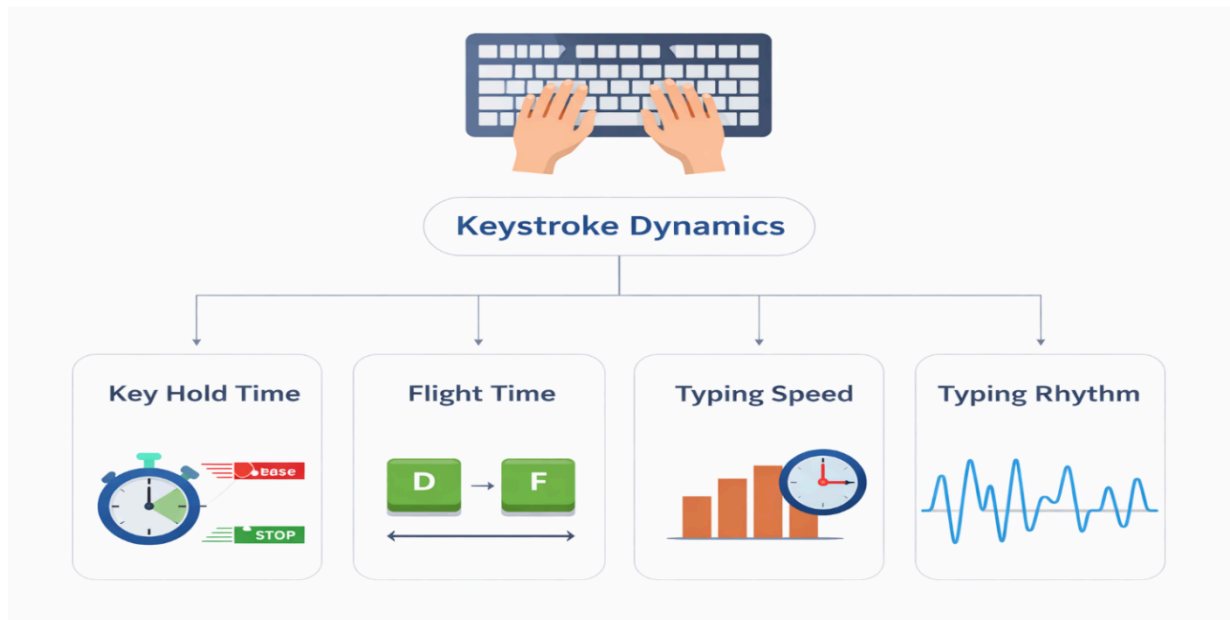


Fig. 3.1 Classification of Behavioral Biometric Authentication Techniques

Behavioral biometric systems can generally be categorized into the following types:

Keystroke Dynamics: This technique analyzes typing patterns such as key hold time, flight time between keystrokes, typing speed, and typing rhythm.

Mouse Dynamics: This technique analyzes mouse interaction behaviour including cursor movement speed, acceleration, click patterns, and trajectory paths.

Hybrid Behavioral Biometrics: Hybrid systems combine multiple behavioral signals such as keyboard and mouse interactions to improve authentication accuracy and robustness.

Hybrid systems generally achieve better performance compared to single-modality systems because they capture multiple behavioral characteristics simultaneously.

3.1.1 Existing System Architecture

Traditional keystroke dynamics authentication systems as discussed in Wang and Hou [1] rely on machine learning techniques to verify user identity by analyzing typing patterns. These systems aim to enhance security beyond traditional password-based authentication by leveraging behavioral biometrics that are unique to each individual.

- The authentication process in existing keystroke dynamics systems typically consists of multiple stages including data collection, preprocessing, feature extraction, model training, and evaluation, as implemented in prior research including the CMU benchmark dataset studies.
- **SMOTE (Synthetic Minority Over-sampling Technique):** The paper uses SMOTE to address the challenge of **class imbalance**, where genuine user samples significantly outnumber impostor samples. It generates synthetic samples for the minority class (impostors) to create a balanced dataset for better model training.

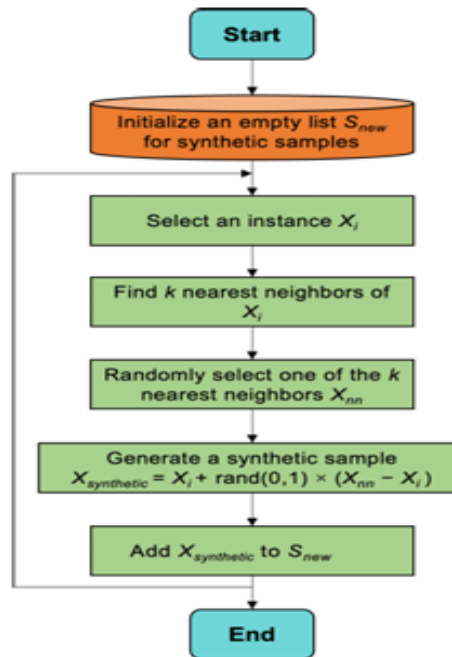


Fig 3.2 SMOTE architecture [1].

- **XGBoost (Extreme Gradient Boosting):** The paper applies XGBoost as one of its primary **ensemble learning** methods. It is used to improve classification performance and robustness compared to traditional machine learning algorithms

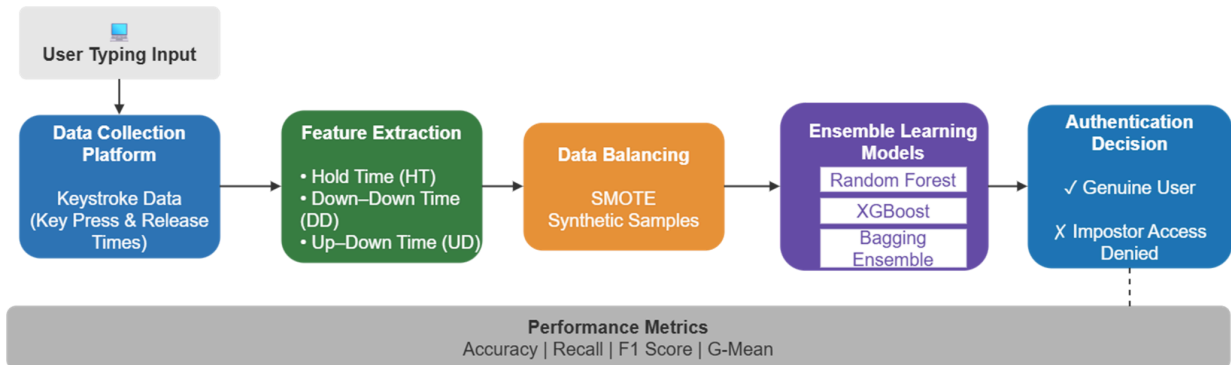


Figure 3.3: Existing Architecture

The authentication process is carried out through the following major components:

• User Interaction Module

This module represents the interaction between the user and the computing system. During system usage, users generate behavioral data through their typing activities on standard keyboards. In existing keystroke dynamics research, users typically input predefined text such as passwords or phrases into the system.

Examples of behavioral signals collected include:

- **Key press events:** Timestamps when keys are pressed down
- **Key release events:** Timestamps when keys are released
- **Key press duration:** The time a key remains pressed (hold time)
- **Time intervals between keystrokes:** Latencies between consecutive key events
- **Typing rhythm:** Overall pattern and speed of typing

According to Wang and Hou [1], these behavioral signals are unique to each individual and can be used to identify users during a session. In datasets like the CMU benchmark, users type the same password multiple times, capturing variations in their typing patterns.

• Data Collection Module

The data collection module captures raw keystroke events as users interact with the system. In existing research, this module records precise timestamps for key press and release actions.

The collected raw data includes:

- Key identifiers (which keys were pressed)
- Press timestamps (when a key is pressed down)
- Release timestamps (when a key is released)
- User identifiers

- Session information

Wang and Hou [1] note that existing datasets in keystroke dynamics research, including the CMU benchmark dataset, are often characterized by:

- **Data scarcity:** Limited number of samples per user
- **Lack of standardized collection methods:** Variations in how data is collected across different studies
- **Small dataset sizes:** Making it difficult to compare results across different studies
- **Unlabelled data:** The original CMU dataset does not contain labels, requiring researchers to generate sub-datasets for classification purposes

• Pre-processing Module

The collected raw behavioral data cannot be used directly for authentication. Therefore, a pre-processing stage is required to convert the raw interaction data into a structured format suitable for analysis.

Typical pre-processing tasks in existing systems include:

- Removing duplicate or noisy keystroke events
- Filtering incomplete typing sequences
- Standardizing key event representations
- Calculating time differences between consecutive keystrokes
- Handling missing or corrupted data points

This stage transforms raw user interaction data into structured datasets suitable for feature extraction and machine learning models. Wang and Hou [1] note that in the CMU dataset, researchers must generate sub-datasets with the same number of subjects for each subject as genuine and the rest as impostors since the original dataset lacks labels.

• Feature Extraction Module

In this stage, meaningful behavioral features are extracted from the processed keystroke interaction data. These features characterize typing behavior and form the basis for user authentication.

Examples of extracted features in existing keystroke dynamics systems include:

- **Key Hold Time (HT):** The duration a key is pressed (release time minus press time)
- **Key Intervals:** Time differences between consecutive keystrokes, including:
 - Up-Down latency (time between releasing one key and pressing the next)
 - Down-Down latency (time between pressing two consecutive keys)
- **Typing speed:** Overall typing rate measured in characters per minute
- **Digraph combinations:** Timing patterns for specific key pairs
- **Flight time:** Time between successive key presses

According to Wang and Hou [1], in traditional keystroke dynamics research, these features are extracted from users typing predefined text, with the key information lying in the varying keystroke characteristics among different users. The feature space can be high-dimensional, particularly when considering timing patterns for multiple key combinations.

● **Traditional Machine Learning Models Module**

This module implements traditional machine learning algorithms that have been widely applied to keystroke dynamics for distinguishing between legitimate users and impostors based on typing patterns. Wang and Hou [1] identify four primary algorithms commonly used in existing systems:

Decision Trees (DT)

Decision Trees are favoured for their simplicity and interpretability. In keystroke dynamics, features such as keypress duration and typing speed are used to build decision rules that help classify users. The tree structure creates if-then rules based on feature values to determine whether typing patterns belong to legitimate users or impostors.

Limitations identified by Wang and Hou [1]:

- Decision Trees are prone to high variance and overfitting
- Particularly problematic when dealing with high-dimensional keystroke data
- Techniques such as pruning are applied but are not always sufficient for complex data

Support Vector Machines (SVM)

SVMs are well-suited for binary classification tasks in keystroke dynamics, distinguishing between genuine users and impostors. They work by identifying an optimal hyperplane that maximizes the margin between different classes. This makes them a strong candidate for handling the complex feature relationships in keystroke dynamics, such as the interaction between keystroke durations and key intervals.

Limitations identified by Wang and Hou [1]:

- SVMs can be computationally expensive, especially with larger datasets
- They often struggle with imbalanced data where one class (typically impostors) is underrepresented
- The class imbalance problem is critical in keystroke dynamics where genuine samples may significantly outnumber impostor samples

k-Nearest Neighbours (k-NN)

k-NN is a simple and intuitive non-parametric method used for classification in keystroke dynamics. It compares a user's typing pattern to the stored patterns of known users and assigns a class based on the majority of the k-nearest neighbours in the feature space.

Limitations identified by Wang and Hou [1]:

- k-NN becomes computationally intensive with larger datasets
- Performs poorly in high-dimensional feature spaces due to the curse of dimensionality
- Keystroke dynamics data often has many features (dimensions) but limited samples

Neural Networks (NN)

Neural Networks capture complex, non-linear patterns in data, making them suitable for keystroke dynamics authentication where intricate relationships between typing behaviors need to be modeled. They can learn hierarchical representations of typing patterns.

• Evaluation Module

The evaluation component determines the authentication decision based on the model's predictions. The system compares the observed behavioral patterns with stored profiles and classifies the user as either genuine or impostor.

Performance of existing authentication systems is evaluated using metrics such as:

- **Accuracy:** Proportion of correctly classified instances out of total instances
- **Recall:** Proportion of actual positive instances (genuine users) correctly identified
- **G-Mean:** Geometric mean of sensitivity and specificity, providing a balanced measure
- **F1-Score:** Harmonic mean of precision and recall

Wang and Hou [1] note that in the CMU dataset, which is inherently imbalanced with a higher proportion of impostor inputs compared to genuine inputs, traditional classifiers tend to predict the majority class (impostor users) more frequently. This can result in high accuracy but poor performance in identifying the minority class (genuine users). Therefore, metrics such as G-mean and recall are more indicative of the model's true ability to identify minority class instances.

Limitations of Existing System

- **Exclusion of Multimodal Data:** The methodology relies strictly on keystroke dynamics, omitting mouse dynamics or other behavioral modalities that significantly enhance authentication robustness
- **Static Adaptability:** The model fails to explicitly address long-term behavior evolution, such as changes in typing speed or proficiency over months or years
- **Limited Longitudinal Scope:** The study lacks analysis of user patterns separated by significant time margins, leaving the "adaptability gap" unresolved
- **Potential for Overfitting:** High-dimensional biometric features combined with small sample sizes create significant risks of overfitting, even when using ensemble learning

3.1.2 Proposed System Architecture

The previous section discussed the limitations of existing behavioral biometric authentication systems as presented in Wang and Hou [1]. Traditional systems rely mainly on supervised

machine learning models (Decision Trees, SVM, k-NN, Neural Networks) trained on imbalanced datasets without addressing class imbalance, leading to high variance, overfitting, and poor detection of minority classes (genuine users). Additionally, these systems lack mechanisms to adapt to dynamic changes in user behaviour over time, further reducing their reliability and accuracy in real-world continuous authentication scenarios.

To overcome these issues, a Secure Adaptive Multimodal Continuous Authentication System is proposed. The proposed architecture integrates behavioral biometric analysis from both keyboard and mouse modalities with adaptive machine learning techniques across five distinct phases. Unlike the existing system described in Wang and Hou [1], which relies solely on traditional classifiers with imbalanced keystroke data, this proposed system incorporates multi-source data collection, feature selection, SMOTE balancing, hybrid model training with both classification and anomaly detection, and continuous adaptation mechanisms.

3.2 Requirements for Implementation

3.2.1 Algorithms / Techniques

Phase 1: Multi-Source Data Collection

Module 1.1: Keyboard Data Capture Module

This module captures raw keystroke events from user interactions with the keyboard. Unlike the existing system in Wang and Hou [1] which relies solely on keystroke data from predefined text inputs, this module records Key Down Timestamps (precise time when each key is pressed), Key Up Timestamps (precise time when each key is released), and Key Identifiers (which keys are being pressed). This raw timestamp data forms the foundation for all subsequent keystroke analysis.

Module 1.2: Mouse Data Capture Module

This module extends beyond the existing system by incorporating mouse dynamics as an additional biometric modality. It captures Mouse X-Y Coordinates (position data at regular intervals), Mouse Event Types (clicks, movements, scrolling), and associated Timestamps. The inclusion of mouse data provides a second behavioral modality, creating a richer biometric profile than keystroke-only systems.

Module 1.3: Sliding Window Segmentation Module

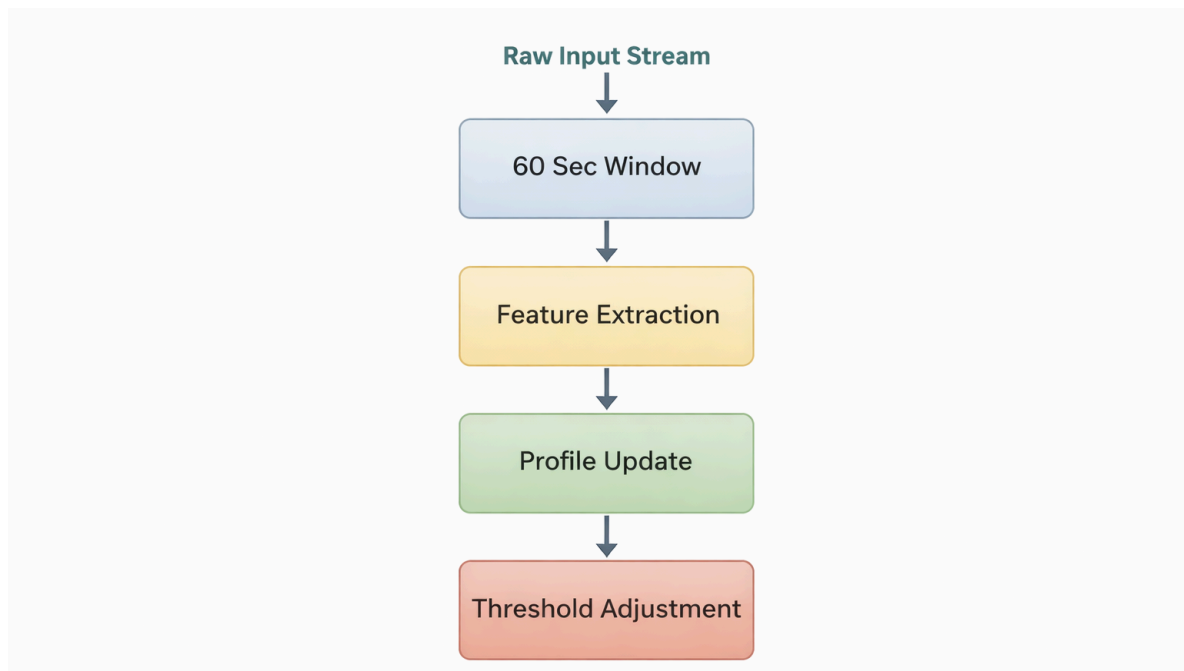


Fig. 3.4 Sliding Window Behaviour Update Mechanism

This module groups continuous data streams into manageable units for analysis. Unlike the existing system which analyses isolated typing events, this module provides Fixed-Time Windows (grouping data into 60-second intervals) and Fixed-Event Sequences (grouping based on number of events). This approach preserves behavioral context rather than analysing isolated actions, enabling more accurate pattern recognition.

Phase 2: Feature Engineering

Module 2.1: Keystroke Feature Extraction Module

This module extracts high-level metrics from raw keystroke data, similar to but more comprehensive than the existing system. It calculates Hold Time (HT) which is the duration a key remains pressed (release time minus press time), Up-Down Latency (UD) representing the time between releasing one key and pressing the next, Down-Down (DD) Digraphs measuring time between pressing consecutive keys, and Typing Rhythm Patterns capturing sequential timing characteristics. These features collectively represent the user's unique typing fingerprint.

Module 2.2: Mouse Feature Extraction Module

This module extracts behavioral features from mouse movements, adding a modality not present in the existing system. It computes Velocity (speed of mouse movements), Trajectory Straightness (how directly the mouse moves between points), Angle of Curvature (smoothness of curved movements), and Clicking Frequency (rate and pattern of mouse clicks). These mouse dynamics features complement keystroke data for more robust authentication.

Module 2.3: mRMR Feature Selection Module

This module addresses the high-dimensional data challenge identified in Wang and Hou [1].

The Minimum Redundancy Maximum Relevance (mRMR) algorithm selects approximately 36 most discriminative features from the combined keystroke and mouse feature set. It minimizes redundancy by removing correlated features while maximizing relevance by retaining features most predictive of user identity. This dimensionality reduction directly addresses the curse of dimensionality that plagued traditional classifiers in the existing system.

Phase 3: Handling Data Imbalance

Module 3.1: SMOTE Resampling Module

This module directly addresses the class imbalance limitation identified in Wang and Hou [1]. The Synthetic Minority Oversampling Technique (SMOTE) resolves the 1:44 ratio between genuine and impostor samples by generating synthetic examples of "not-you" behaviour. For each minority class sample, SMOTE selects k nearest neighbours from the same class and creates new synthetic samples through interpolation between existing samples:

Unlike simple duplication, SMOTE preserves data diversity, enabling models to learn clear decision boundaries even with limited physical impostor data.

Phase 4: Hybrid Model Training

Module 4.1: LightGBM Classifier Module - "The Discriminator"

This module provides supervised classification capability that improves upon the traditional classifiers in Wang and Hou [1]. LightGBM (Light Gradient Boosting Machine) is a gradient boosting framework that uses tree-based learning algorithms. It offers fast training speed due to its histogram-based algorithms and leaf-wise growth strategy. The model achieves high classification accuracy of 94.68% in distinguishing between genuine users and impostors. As an ensemble method, it combines multiple weak learners to create a strong classifier, reducing the variance problem that affected traditional Decision Trees in the existing system. The classifier is trained on the SMOTE-balanced dataset, ensuring it learns effectively from both classes.

Module 4.2: One-Class SVM Module - "The Gatekeeper"

This module adds anomaly detection capability not present in the existing system. The One-Class Support Vector Machine is trained exclusively on normal user data, learning the boundaries of legitimate user behaviour without requiring any impostor labels. It creates a hypersphere or hyperplane that encapsulates normal behaviour patterns. During authentication, it flags any behavior that falls outside these learned boundaries as statistically significant deviations. This provides complementary security by detecting novel attacks that the supervised classifier hasn't seen during training, addressing the limitation of traditional systems that require impostor examples for training.

Module 4.3: SHAP Explainability Module

This module adds transparency to the authentication process, a feature completely absent in the existing system. SHAP (SHapley Additive exPlanations) values rank feature importance based on cooperative game theory, identifying exactly which behavioral markers are most vital to each authentication decision. For example, while LightGBM may confirm the typing speed matches the user, SHAP can reveal if Hold Time variance is unusually high due to stress or shaking hands from coercion. This explainability ensures that authentication decisions are transparent, auditable, and interpretable, building trust in the system.

Phase 5: Continuous Authentication & Adaptivity

Module 5.1: Real-Time Scoring Module

This module evaluates every new sliding window using both trained models. It computes classification scores from LightGBM and anomaly scores from One-Class SVM for each window of user activity. The continuous monitoring operates in the background without interrupting user experience, achieving the "seamless background authentication" that Wang and Hou [1] identified as a key advantage of behavioral biometrics.

Module 5.2: Threshold-Based Decision Module

This module determines authentication actions based on the computed anomaly scores. It compares the anomaly score against predefined thresholds that balance security and usability. If the anomaly score is less than or equal to the threshold, the user is considered legitimate and access is granted with the session continuing normally. If the anomaly score exceeds the threshold, the system identifies suspicious behaviour and triggers the re-authentication process.

Module 5.3: Re-authentication Trigger Module

This module initiates additional verification when anomalies are detected. It sends a One-Time Password (OTP) request to the user's registered device, requiring step-up authentication before allowing sensitive operations to continue. This provides an additional security layer while maintaining the primary continuous authentication in the background.

Module 5.4: Adaptive Profile Update Module

This module addresses the critical limitation of static behavioral profiles in existing systems. It implements a sliding window approach to profile maintenance where new high-confidence samples (windows where authentication was successful and confidence was high) are continuously added to the user's behavioral profile. Simultaneously, the oldest data points are retired from the profile. This mechanism accommodates long-term behavioral changes such as typing speed improving over years, adaptation to new keyboards, or changes due to aging or health conditions. The adaptive update ensures the system remains accurate throughout the

user's lifetime.

Module 5.5: Feedback Loop Module

This module completes the continuous authentication cycle by periodically retraining models with the updated user profiles. It incorporates new behavioral patterns into the LightGBM classifier and updates the One-Class SVM boundary based on evolved normal behavior. This creates a self-improving system that becomes more accurate and robust over time, continuously adapting to the user's evolving behavioral characteristics.

3.2.2 Use Case Diagram / Activity Diagram / DFDs.

The proposed architecture is shown in Figure 3.3

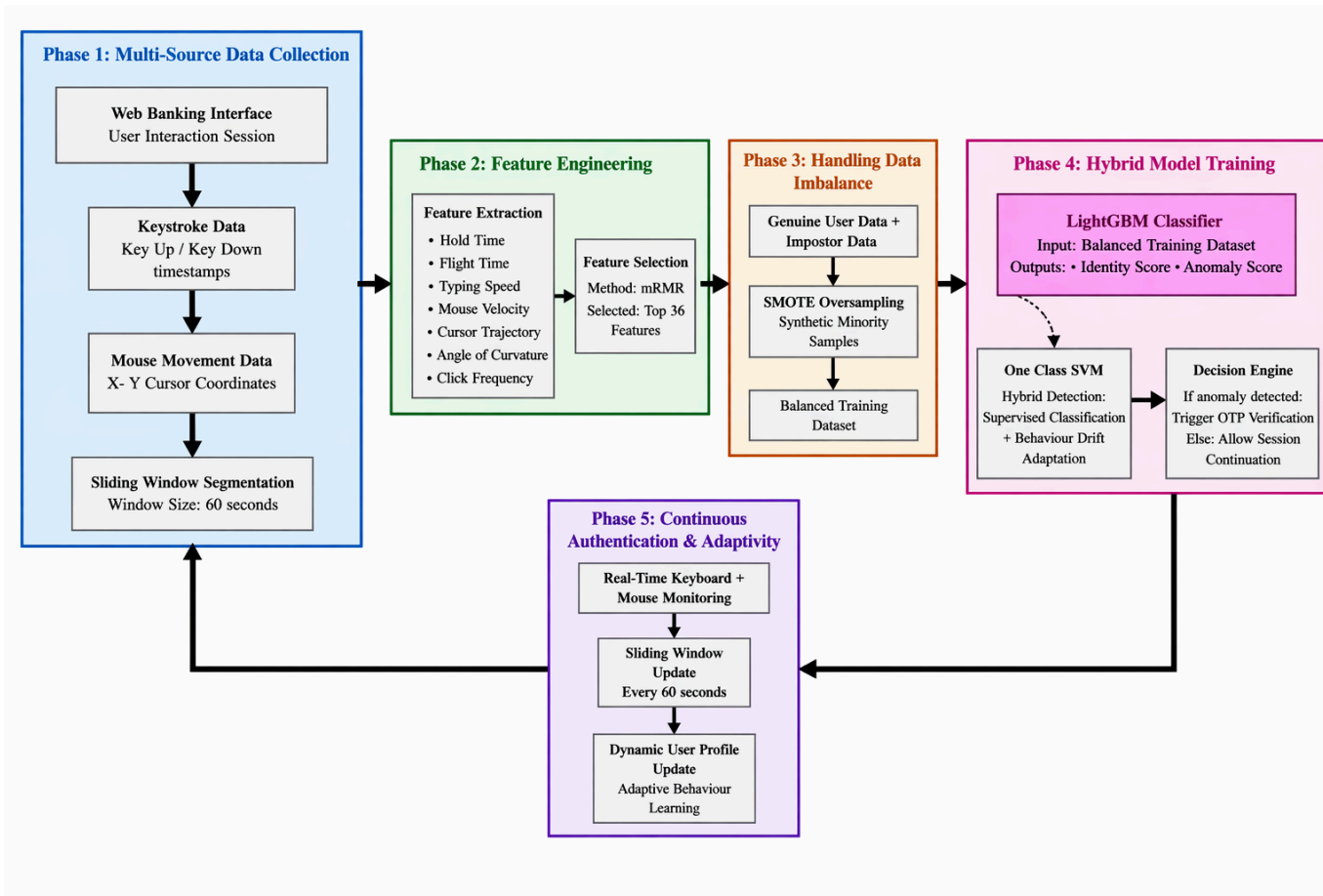


Fig. 3.5 Proposed Continuous Authentication System Architecture

3.2.3 Sample Dataset Used

An experiment is conducted to analyze the behaviour of the authentication system. The dataset contains behavioral biometric data collected from users interacting with input devices.

Table 3.1 shows the datasets used for the experiment.

Table 3.1 Sample Dataset Used for Experiment

Dataset	Users	Features	Sessions	Type
CMU Keystroke Dataset	51	31	20,400	Keystroke
Buffalo Free Text Dataset	148	Multiple	Continuou s	Keystroke
Custom Behavioral Dataset	20–3 0	~120	Multiple	Keyboard + Mouse

These datasets contain behavioral interaction patterns used for training and testing the authentication system.

3.2.4 Hardware and Software Specifications

The experimental setup is carried out using the following hardware and software configuration.

Table 3.2 Hardware Details

Component	Specification
Processor	Intel Core i5
Storage	512 GB SSD
RAM	8 GB

Table 3.3 Software Details

Component	Specification
Operating System	Windows 10 / Linux
Programming Language	Python
Libraries	Scikit-learn, NumPy, Pandas
Framework	Flask
Database	MongoDB

3.3 Evaluation Metrics

The performance of the authentication system is evaluated using biometric authentication metrics.

Precision

Precision measures the proportion of correctly authenticated users among all users classified as legitimate.

$$Precision = \frac{True\ Positives}{True\ Positives + False\ Positives}$$

Recall

Recall measures the proportion of legitimate users correctly identified by the system.

$$Recall = \frac{True\ Positives}{True\ Positives + False\ Negatives}$$

Accuracy

Accuracy represents the overall correctness of the authentication system.

Equal Error Rate (EER)

EER represents the point where False Acceptance Rate (FAR) equals False Rejection Rate (FRR). A lower EER indicates better system performance.

Chapter 4

Applications

Continuous behavioral biometric authentication systems are increasingly used in modern digital environments to enhance security beyond traditional login-based verification. Unlike static authentication methods, continuous authentication verifies user identity throughout the session using behavioral characteristics such as typing rhythm, mouse movement patterns, and touchscreen interactions. These systems provide an additional security layer across multiple domains. The major social and technical applications of the proposed system are discussed below.

4.1 Social

Continuous authentication systems improve digital safety for individuals by protecting personal information and preventing identity misuse in everyday online activities.

1. Online Banking Security

Online banking platforms manage highly sensitive financial data and transactions. Traditional authentication methods such as passwords and OTP verification provide only one-time access control during login.

Continuous authentication enhances security by:

- monitoring typing behavior during transactions
- detecting abnormal interaction patterns
- preventing unauthorized session continuation after login

Even if login credentials are compromised, the system can identify suspicious behavioral deviations and terminate the session automatically.

2. E-Commerce Platform Protection

E-commerce applications store customer profiles, payment details, and transaction histories, making them attractive targets for attackers.

Continuous authentication helps by:

- detecting fraudulent purchase behavior
- identifying suspicious login activity
- preventing account takeover attacks

Behavioral monitoring ensures that unauthorized users cannot continue sessions even after credential theft.

3. Secure Access to Social Media Platforms

Social media platforms store personal conversations, images, and sensitive user-generated content.

Continuous authentication systems help:

- detect unauthorized access attempts
- prevent misuse of personal accounts
- monitor abnormal browsing patterns

This improves privacy protection for users in digital communication environments.

4. Mobile Device Security

Smartphones and personal devices store personal information such as messages, contacts, and banking applications.

Behavioral authentication enhances device protection by:

- analyzing touchscreen interaction patterns
- monitoring gesture-based activity
- detecting abnormal device usage behavior

This allows continuous verification without requiring repeated password entry.

4.2 Technical Applications

Continuous authentication systems play a critical role in strengthening cybersecurity infrastructure across enterprise networks and intelligent digital platforms.

1. Enterprise Network Security

Corporate systems contain confidential organizational data and intellectual property assets.

Continuous authentication supports:

- monitoring employee interaction with internal systems
- detecting unauthorized database access
- preventing insider threat activities

This improves protection of enterprise infrastructure and sensitive corporate information.

2. Secure Web Applications

Modern web applications store important personal and professional information including emails, documents, and cloud storage data.

Continuous authentication helps:

- monitor session-level user activity
- detect suspicious behavior patterns
- prevent unauthorized access to stored resources

This ensures secure access control throughout active user sessions.

3. Fraud Detection Systems

Behavioral biometrics are widely applied in financial fraud detection systems because each

user exhibits unique interaction characteristics.

Applications include:

- detecting abnormal transaction behavior
- identifying identity theft attempts
- recognizing suspicious login sequences

These systems significantly reduce financial losses caused by cybercrime.

4. Cybersecurity and Intrusion Detection Systems

Continuous authentication can be integrated with cybersecurity monitoring frameworks to strengthen threat detection capabilities.

It supports:

- identifying unauthorized system access
- monitoring behavioral deviations in real time
- detecting automated attack tools and bot activities

This provides an additional defense layer beyond conventional authentication mechanisms.

Chapter 5

Summary

In this report, the study of different techniques used in continuous authentication systems based on behavioral biometrics has been presented. The system analyses user behavioral patterns such as typing rhythm and mouse movements to continuously verify user identity during system interaction.

Traditional authentication methods such as passwords and PINs provide only one-time verification and are vulnerable to attacks such as credential theft and session hijacking. To overcome these limitations, behavioral biometric techniques such as keystroke dynamics and mouse dynamics are used to provide continuous authentication.

The proposed Secure Adaptive Multimodal Continuous Authentication System combines behavioral data collection, feature extraction, and machine learning models to authenticate users based on their interaction patterns. The system continuously monitors user behavior and detects anomalies that may indicate unauthorized access.

Different performance metrics such as accuracy, precision, recall, False Acceptance Rate (FAR), and False Rejection Rate (FRR) are used to evaluate the effectiveness of the authentication system.

Various datasets used for behavioral biometric research are also discussed, and the potential applications of this system in domains such as online banking, enterprise security, e-commerce, mobile devices, and cybersecurity systems are identified.

Continuous authentication systems provide a strong additional layer of security by verifying user identity throughout the session, making them a promising solution for protecting modern digital systems from unauthorized access.

References

- [1] X. Wang and D. Hou, “Enhancing keystroke dynamics authentication with ensemble learning and data resampling techniques,” *Electronics*, vol. 13, no. 22, Art. no. 4559, Nov. 2024.
- [2] A. Muralidharan, A. Eaman, and E. Hassan, “A comparative analysis of machine learning models for behavioral biometric authentication using keystroke dynamics,” *Procedia Computer Science*, vol. 265, pp. 116–123, 2025.
- [3] A. Subash, I. Song, I. Lee, and K. Lee, “Adaptability of current keystroke and mouse behavioral biometric systems: A survey,” *Computers & Security*, vol. 160, p. 104731, 2026.
- [4] A. Almeahmadi, “Cyber coercion detection using LLM-assisted multimodal biometric system,” *Applied Sciences*, vol. 15, no. 19, Art. no. 10658, Oct. 2025.
- [5] A. Martins, T. Dias, A. Dias, J. Vitorino, E. Maia, and I. Praça, “Keystroke dynamics for intelligent biometric authentication with machine learning,” *Discover Applied Sciences*, vol. 7, Art. no. 992, 2025.
- [6] K. Asgarov, “Endpoint behavior modelling from user interaction metrics via unsupervised deep learning for anomaly detection,” Azerbaijan Technical University, 2025.
- [7] Ł. Wyciślik, P. Wylężek, and A. Momot, “The improved biometric identification of keystroke dynamics based on deep learning approaches,” *Sensors*, vol. 24, no. 12, Art. no. 3763, Jun. 2024.
- [8] P. Bansal and A. Ouda, “Continuous authentication in the digital age: An analysis of reinforcement learning and behavioral biometrics,” *Computers*, vol. 13, no. 4, Art. no. 103, Apr. 2024.
- [9] L. Yang and S. Qin, “Identify user features impacting keystroke, mouse and touchscreen dynamics,” *Multimedia Tools and Applications*, vol. 84, pp. 48685–48713, 2025.
- [10] U. Uslu, O. D. Incel, and G. I. Alptekin, “Evaluation of deep learning models for continuous authentication using behavioral biometrics,” *Procedia Computer Science*, vol. 225, pp. 1272–1281, 2023.
- [11] X. Wang *et al.*, “User authentication method based on keystroke dynamics and mouse dynamics with scene-irrelated features in hybrid scenes,” *Sensors*, vol. 22, no. 17, Art. no. 6627, Sept. 2022.
- [12] S. Krishnamoorthy, L. Rueda, S. Saad, and H. Elmiligi, “Identification of user behavioral biometrics for authentication using keystroke dynamics and machine learning,” in *Proc. ICBEA*, 2018, pp. 50–57.
- [13] A. F. Baig, S. Eskeland, and B. Yang, “Privacy-preserving continuous authentication using behavioral biometrics,” *International Journal of Information Security*, vol. 22, pp. 1833–1847, 2023.
- [14] J. Zhang, Y. Wang, and L. Li, “SoundAuth: Non-intrusive user authentication via keystroke acoustic biometrics with triplet networks,” University of Electronic Science and Technology of China, 2025.

- [15] Z. Jorgensen and T. Yu, “On mouse dynamics as a behavioral biometric for authentication,” in *Proc. ASIA CCS*, 2011, pp. 475–482.
- [16] M. Abuhamad, A. Abusnaina, D. Nyang, and D. Mohaisen, “Sensor-based continuous authentication of smartphone users: A contemporary survey,” *arXiv preprint arXiv:2001.08578*, 2020.
- [17] R. Shadman, A. Wahab, M. Manno, M. Lukaszewski, D. Hou, and F. Hussain, “Keystroke dynamics: Concepts, techniques, and applications,” *ACM Computing Surveys*, vol. 57, no. 11, 2025.
- [18] P. Kasprowski, Z. Borowska, and K. Harezlak, “Biometric identification based on keystroke dynamics,” *Sensors*, vol. 22, no. 9, Art. no. 3158, Apr. 2022.

A Comprehensive Survey on Adaptive Continuous User Authentication Using Behavioral Biometrics

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Abstract—

Continuous authentication using behavioral biometrics has emerged as a transformative approach to cybersecurity, enabling identity verification throughout active user sessions rather than only at the point of login. This survey provides a comprehensive analysis of keystroke dynamics, mouse movements, touchscreen gestures, and sensor-based behavioral signals, along with the machine learning and deep learning techniques employed for identity verification. We examine critical challenges including behavioral drift over time, contextual variability across different tasks and environments, device heterogeneity that affects signal consistency, and privacy preservation requirements for real-world deployment. Based on this analysis, we propose a hybrid adaptive authentication architecture that integrates sliding window segmentation, multimodal feature extraction, LightGBM classification, One-Class SVM anomaly detection, and incremental profile updating mechanisms. This architecture is designed to improve long-term usability, robustness against behavioral evolution, and practical deployment capability across diverse application domains including online banking, enterprise security, and mobile platforms.

IndexTerms— Behavioral biometrics, continuous authentication, keystroke dynamics, mouse dynamics, adaptive authentication, deep learning, cybersecurity, multimodal fusion, behavioral drift.

I. INTRODUCTION

Traditional authentication systems, including password-based verification, PIN entry, and even single-factor biometric systems, verify user identity only once at the point of login. While these methods have served as the cornerstone of digital security for decades, they suffer from a fundamental vulnerability: once the initial authentication is complete, the system has no

mechanism to verify that the same legitimate user remains in control of the session. This limitation creates significant security risks including session hijacking attacks, where an attacker takes over an already-authenticated session; insider threats, where a legitimate user's credentials are co-opted by malicious actors; and unauthorized access following device theft after login [7], [13]. The financial impact of these vulnerabilities is substantial, with identity theft losses reaching approximately USD 43 billion annually according to recent industry reports, and phishing attacks increasing by 75% over a two-year period to reach nearly 5 million detected instances.

Continuous authentication systems address this fundamental limitation by monitoring behavioral interaction patterns throughout active sessions rather than relying on a single verification event. These systems collect and analyze how users interact with their devices in real-time, examining characteristics such as typing rhythm (how quickly and with what patterns users press keys), mouse movement trajectories (the paths cursors follow across screens), touchscreen gestures (swipe patterns, tap pressure, and finger contact area), and motion sensor signals (accelerometer and gyroscope readings that capture how users hold and move their devices). The fundamental premise underlying behavioral biometrics is that each individual possesses distinctive, consistent, and difficult-to-replicate patterns of interaction that can serve as a reliable identifier[2], [8]. Unlike passwords that can be stolen or shared, behavioral characteristics are inherently tied to the user and operate transparently without requiring any additional action[11], [13].

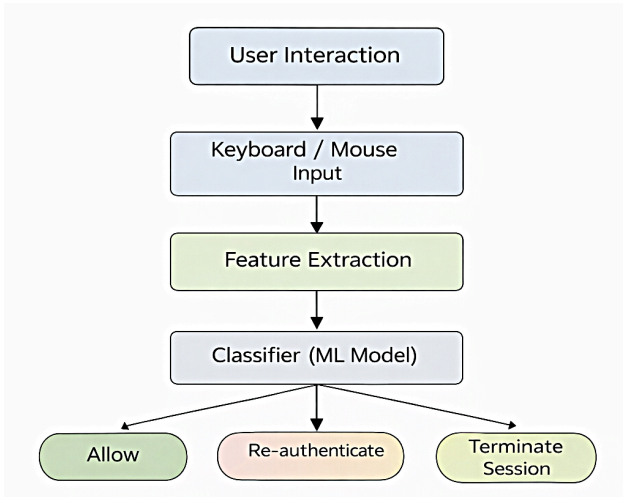


Fig. 1. Continuous behavioral biometric authentication workflow using keyboard and mouse interaction signals.

Behavioral biometrics offers several compelling advantages over traditional authentication methods and even over physiological biometrics such as fingerprints or facial recognition. First, behavioral data can be collected non-intrusively using standard input devices that are already present on computers and mobile devices, eliminating the need for specialized hardware such as fingerprint scanners or iris cameras. Second, authentication occurs transparently in the background without interrupting user workflow, providing security without compromising usability or productivity. Third, behavioral patterns are significantly more difficult to forge or replicate than static credentials, as they involve dynamic characteristics that vary naturally across individuals. Fourth, behavioral biometrics enables continuous monitoring throughout sessions, detecting anomalies even after initial authentication has been performed[5], [8].

Despite these advantages and significant research progress over the past decade, several critical challenges continue to hinder widespread real-world deployment of behavioral biometric authentication systems. Behavioral drift occurs as user interaction patterns naturally evolve over time due to fatigue, skill acquisition through practice, emotional state changes, illness, aging, and adaptation to new devices or software interfaces. A user who types slowly and deliberately when first learning a system may type rapidly and automatically after months of regular use, causing authentication systems trained on early data to incorrectly reject the legitimate user[13], [21]. Contextual variability refers to the observation that the same user exhibits different behavioral patterns when performing different tasks, using different devices, or working in different environments. A user's mouse movements during precise graphic design work differ substantially from their

movements during casual web browsing, and systems that do not account for these contextual differences may generate false rejections[10], [22]. Device heterogeneity means that behavioral signals captured from different keyboards, mice, touchscreens, or smartphones are not directly comparable; typing on a mechanical desktop keyboard produces different timing characteristics than typing on a laptop keyboard or a touchscreen, and authentication systems must handle this variability[8], [21].

Privacy preservation presents another significant challenge, as continuous authentication systems necessarily collect sensitive behavioral data that could potentially reveal personal information about users including their activity patterns, emotional states, health conditions, and even demographic characteristics. Users may be uncomfortable with continuous monitoring, and regulatory frameworks such as GDPR impose strict requirements on the collection, storage, and processing of behavioral data[12]. Finally, the lack of standardized, publicly available long-term datasets spanning months or years makes it difficult to fairly compare different authentication approaches and evaluate their performance under realistic conditions where behavioral drift occurs[7], [13].

This survey provides a comprehensive analysis of the current state of behavioral biometric-based continuous authentication systems. Section II establishes background and fundamental concepts including the distinction between static and continuous authentication, the range of behavioral modalities, and key evaluation metrics. Section III examines keystroke dynamics in depth, covering datasets, feature extraction, and classification approaches from traditional machine learning to deep learning. Section IV explores mouse dynamics and multimodal systems that combine multiple behavioral signals. Section V addresses mobile sensor-based authentication using accelerometers, gyroscopes, and touchscreen data. Section VI examines deep learning approaches including CNN, LSTM, and transformer architectures. Section VII covers privacy-preserving authentication techniques including homomorphic encryption and secure template storage. Section VIII addresses the critical challenge of behavioral drift and adaptive authentication mechanisms. Section IX presents our proposed hybrid adaptive authentication architecture designed to address identified gaps. Section X concludes the survey with recommendations for future research.

II. BACKGROUND AND FUNDAMENTALS OF CONTINUOUS AUTHENTICATION

Continuous authentication represents a paradigm shift from traditional static authentication approaches by verifying user identity repeatedly and transparently throughout an active session rather than only at login time. Unlike conventional authentication mechanisms that rely on passwords, tokens, or one-time biometric verification, continuous authentication systems analyze behavioral interaction patterns dynamically during system usage to ensure that the legitimate user remains in control of the session [8], [13].

Traditional static authentication assumes that the entity authenticated at login remains unchanged throughout the session duration. However, this assumption often fails in real-world environments due to session hijacking attacks, insider threats, and unauthorized access following credential compromise. Continuous authentication addresses these vulnerabilities by monitoring user interaction characteristics such as typing rhythm, mouse movement trajectories, touchscreen gestures, and sensor-based motion signals in real time [7], [22].

Behavioral biometric authentication systems operate transparently in the background without interrupting user workflows, providing a balance between security and usability. Since behavioral interaction patterns are difficult to imitate or reproduce accurately, they offer a reliable mechanism for detecting unauthorized users even after successful login authentication [2], [11].

Despite their advantages, continuous authentication systems must address several challenges including behavioral drift, contextual variability, device heterogeneity, and privacy concerns. These challenges significantly influence the long-term reliability and deployment feasibility of behavioral biometric authentication systems in practical environments [21], [13].

A. Static Versus Continuous Authentication

Static authentication, which encompasses the vast majority of deployed authentication systems today, operates on a simple model: the user provides credentials at the beginning of a session, the system verifies these credentials against stored references, and if verification succeeds, the user is granted access for the duration of the session. This model implicitly assumes that the individual who successfully authenticated remains in control of the session throughout its entire duration. In practice, this assumption is frequently violated. Session hijacking attacks allow attackers to take over authenticated sessions by stealing session cookies or tokens. Insider threats involve legitimate users whose credentials are co-opted by

malicious actors. Device theft after authentication gives attackers direct access to authenticated systems. Credential theft through phishing or keylogging allows attackers to authenticate as legitimate users from different locations. In all these scenarios, static authentication provides no protection because the system has no mechanism to detect that the controlling entity has changed[8], [13].

Continuous authentication addresses this fundamental weakness by performing identity verification repeatedly throughout the session. Rather than relying on a single verification event, continuous authentication systems collect and analyze behavioral data as users naturally interact with their devices[7],[8]. The authentication process operates in the background without requiring explicit user action, extracting features that capture distinctive behavioral patterns and comparing these patterns against stored templates representing the legitimate user's expected behavior[2], [11]. When the observed behavior deviates sufficiently from the expected pattern, the system can take appropriate action ranging from subtle re-authentication challenges (such as requesting the user to re-enter a password or provide a fingerprint) to complete session termination.

The frequency of verification in continuous authentication systems can be adjusted based on risk level[8]. During high-risk activities such as financial transactions or access to sensitive data, more frequent verification can be performed. During low-risk activities such as browsing public information, less frequent verification may suffice. The authentication decision can also incorporate multiple factors: a single anomalous action may not trigger a response, but a sustained pattern of anomalous behavior will.

Several important advantages distinguish continuous authentication from static approaches. First, continuous authentication significantly reduces the attack window available to adversaries who gain access after initial authentication. Research has demonstrated that while an intruder can perform more than 1,000 malicious actions after gaining access to a mobile device using only knowledge-based authentication, multimodal behavioral biometrics reduces this to nearly zero successful actions[5], [8]. Second, continuous authentication provides protection against session hijacking and insider threats that static authentication cannot address. Third, because behavioral biometrics operate transparently, continuous authentication can be implemented without degrading user experience or requiring users to remember additional credentials. Fourth, behavioral characteristics are inherently more difficult to steal or forge than

passwords or physical tokens[13], as they involve dynamic patterns unique to each individual.

However, continuous authentication also presents unique challenges. The requirement for continuous data collection and processing introduces computational overhead and potential battery consumption concerns on mobile devices[7]. The need to make rapid authentication decisions with limited data requires efficient algorithms that can operate reliably with small sample sizes. The inherent variability of human behavior means that authentication decisions must be probabilistic rather than deterministic[11], requiring careful threshold selection to balance security and usability. Finally, the continuous collection of behavioral data raises privacy concerns that must be addressed through appropriate technical and policy measures[12].

B. Behavioral Biometric Modalities

Behavioral biometrics encompasses a diverse set of modalities, each capturing different aspects of human-computer interaction. The most extensively studied modalities in the context of continuous authentication include keystroke dynamics, mouse dynamics, touchscreen gestures, gait analysis, and various sensor-based signals[2], [8].

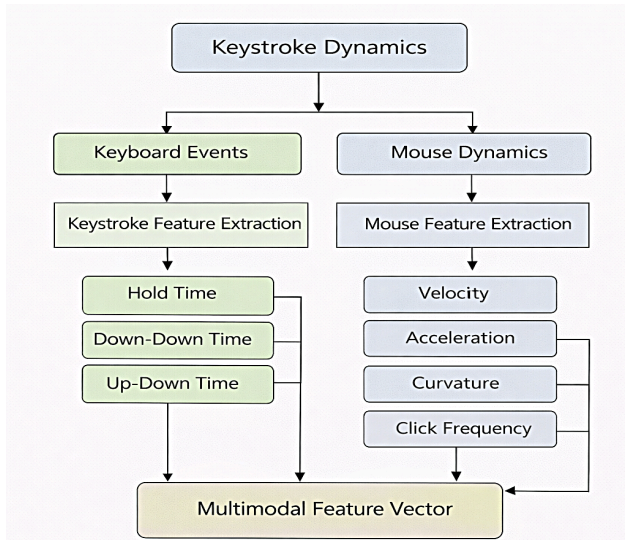


Fig. 2. Multimodal behavioral feature extraction using keystroke timing and mouse trajectory characteristics.

Keystroke dynamics analyzes the timing patterns associated with keyboard typing. When a user types on any keyboard, each key press and release generates precise timing events that can be captured by the operating system or by dedicated logging software. From these raw events, researchers extract various timing features including hold time (the duration between pressing and releasing a single key), flight time (the

interval between releasing one key and pressing the next), down-down time (the interval between pressing consecutive keys), and up-up time (the interval between releasing consecutive keys)[9], [13]. Additional features include typing speed measured in words per minute[6], error rates indicated by backspace and delete key usage, and patterns of special key usage such as Shift, Control, and Alt keys. Keystroke dynamics offers the advantage of being universally applicable across any device with a keyboard, from desktop computers to smartphones with virtual keyboards, and can be collected without specialized hardware. The rich temporal information captured by keystroke timing has been shown to provide strong discriminative power for user identification[11].

Mouse dynamics analyzes the patterns of mouse movement and clicking behavior[1],[10]. When a user operates a mouse, touchpad, or other pointing device, the system captures a continuous stream of events including movement coordinates, button down and up events, and scroll actions. From this raw data, researchers segment the interaction into meaningful units such as point-and-click actions (sequences of movement ending in a click), drag-and-drop actions, and pure movement sequences. Features extracted from these segments include movement speed, acceleration, jerk (rate of change of acceleration)[10], curvature, straightness (ratio of direct distance to actual path length), angle of movement, clicking speed, and clicking strength where measurable. Mouse dynamics is particularly valuable in contexts where typing is infrequent, such as during web browsing, gaming, or interaction with graphical applications, and can serve as a complementary modality to keystroke dynamics in multimodal systems[11].

Touchscreen gestures represent the mobile and tablet analog of mouse dynamics, analyzing the patterns of finger interaction with capacitive touchscreens[7], [8]. Touchscreen data provides richer information than mouse data in several respects, as modern touchscreens can capture not only position and timing but also pressure (the force applied to the screen), area of touch (the size of the finger contact region, which varies with finger orientation and applied force), and multi-touch gestures involving multiple simultaneous finger contacts. Common gestures analyzed for authentication include swipes in various directions, taps, flicks, pinches, spreads, and various custom gestures defined for specific applications[5]. Features extracted from these gestures include velocity, acceleration, path deviation, pressure variation, touch area variation, and gesture duration.

Gait analysis identifies individuals based on their walking patterns using motion sensors embedded in smartphones

and wearables[7], [8]. When a user walks while carrying a mobile device, the device's accelerometer and gyroscope capture characteristic patterns corresponding to the user's stride length, cadence, foot placement, and overall movement dynamics. Gait-based authentication has the unique advantage of operating when the user is not actively interacting with the device, enabling authentication even when the device is in a pocket or bag. This makes gait particularly valuable for continuous authentication in mobile scenarios[5].

Sensor-based signals encompass a wide range of additional modalities enabled by the diverse sensors found on modern mobile devices[7]. These include accelerometer signals capturing device movement and orientation, gyroscope signals capturing angular rotation, magnetometer signals capturing heading and orientation relative to Earth's magnetic field, pressure sensors, proximity sensors, ambient light sensors, and GPS location data. Researchers have explored various combinations of these sensors for authentication, often in conjunction with other modalities[8], [21].

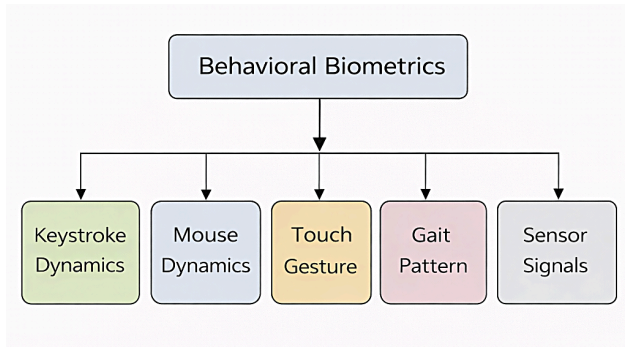


Fig. 3. Classification of behavioral biometric modalities used in continuous authentication systems.

C. Evaluation Metrics for Authentication Systems

The performance of biometric authentication systems is evaluated using several standard metrics that capture different aspects of system behavior[8],[13]. The False Acceptance Rate (FAR) measures the proportion of authentication attempts by impostors that are incorrectly accepted as genuine, calculated as $FP/(FP+TN)$ where FP represents false positives and TN represents true negatives. A low FAR indicates that the system is effective at rejecting impostors, which is critical for security[13]. The False Rejection Rate (FRR) measures the proportion of authentication attempts by genuine users that are incorrectly rejected, calculated as $FN/(FN+TP)$ where FN represents false negatives and TP represents true positives. A low FRR indicates that the system is effective at accepting legitimate users, which is critical for usability. FAR and FRR represent a fundamental trade-off:

tightening the decision threshold to reduce FAR tends to increase FRR, while loosening the threshold to reduce FRR tends to increase FAR.

The Equal Error Rate (EER) is a single-value metric representing the point at which FAR and FRR are equal. The EER is widely used because it provides a threshold-independent summary of system performance, with lower EER values indicating better overall performance[9]. Accuracy measures the overall proportion of correct classifications, calculated as $(TP+TN)/(TP+TN+FP+FN)$. While intuitive, accuracy can be misleading when class distributions are imbalanced. The Receiver Operating Characteristic (ROC) curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at various threshold settings, providing a comprehensive view of classifier performance across the entire operating range[11]. The Area Under the ROC Curve (AUC) provides a single-value summary of ROC performance, with values closer to 1 indicating better discriminative ability[11].

III. KEYSTROKE DYNAMICS

Keystroke dynamics represents the most extensively studied behavioral biometric modality for continuous authentication[2], [13], with research dating back to the pioneering work of Gaines and colleagues in 1980[13]. The fundamental premise is that each individual possesses a distinctive and consistent typing rhythm that can serve as a reliable identifier[6], [9]. This section provides a comprehensive examination of keystroke dynamics-based authentication systems.

A. Types of Keystroke Dynamics Datasets

Keystroke dynamics datasets are generally categorized into static (fixed-text) datasets and dynamic (free-text) datasets[13]. Static datasets collect keystroke data as users type a predetermined text of fixed length and content, typically the same password or phrase across all users and sessions. These datasets are primarily used to evaluate authentication scenarios where the system has advance knowledge of the text being typed[2], such as password-based authentication during login. The CMU Benchmark Dataset[20], collected by Killourhy and Maxion in 2009, is the most widely used static dataset, containing keystroke data from 51 participants who typed the password "tie5Roanl" 50 times per session across 8 sessions, yielding 20,400 samples total[20]. The participant population consisted of 30 males and 21 females with ages ranging from 18-20 to 61-70.

Dynamic datasets collect keystroke data as users engage in unrestricted typing activities that reflect real-world

scenarios[13]. These datasets enable the evaluation of continuous authentication systems that must operate without knowledge of the text being typed. The Buffalo Dataset[9], [15], collected by Sun, Ceker, and Upadhyaya in 2016, contains keystroke data from 148 participants collected over three sessions spanning 28 days, including both transcribed text and free-text responses. The Clarkson II Dataset, collected by Murphy and colleagues in 2017, contains keystroke data from 103 participants typing on desktop keyboards over an extended period of 2.5 years in completely uncontrolled, naturalistic scenarios, making it uniquely valuable for studying long-term behavioral drift[13], [21].

B. Feature Extraction for Keystroke Dynamics

Feature extraction is a critical step in keystroke dynamics-based authentication systems. From raw keystroke events, researchers extract various timing features[6], [13]. Hold time (HT) for a single key is calculated as the difference between release time and press time, capturing how long a user holds down each key[6]. Down-down time (DD) between consecutive keys is calculated as the difference between the press time of the subsequent key and the press time of the current key[6], capturing the overall rhythm of typing. Up-down time (UD), also known as flight time or latency, is calculated as the difference between the press time of the subsequent key and the release time of the current key[6], specifically capturing the gap between keys when no key is pressed. Additional features include typing speed measured in words per minute[9], error rate measured by backspace and delete key frequency, and aggregate statistics including mean, median, standard deviation[13], minimum, and maximum of basic timing features across a typing session

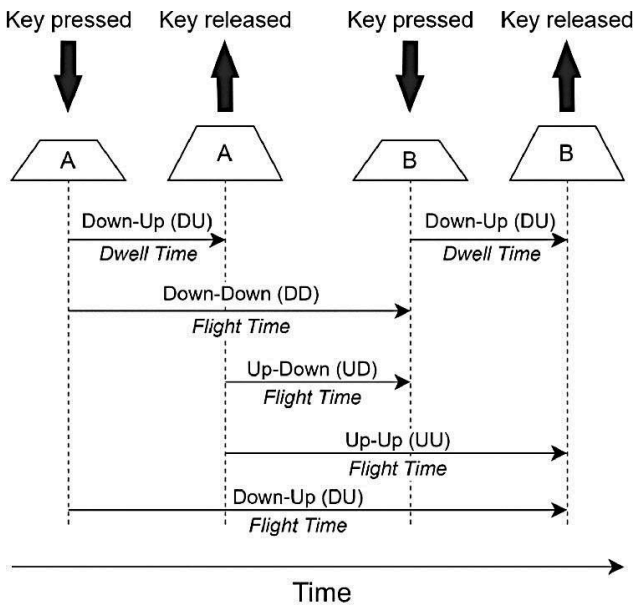


Fig. 4. Keystroke timing feature extraction process showing hold time and inter-key latency measurements..

In addition to feature extraction, feature selection plays a crucial role in reducing dimensionality and improving classifier performance. The Minimum Redundancy Maximum Relevance (mRMR) algorithm is particularly effective for behavioral biometrics, as it selects features that have high correlation with the user identity (maximum relevance) while maintaining low correlation among selected features (minimum redundancy). For a typical keystroke and mouse dynamics system, mRMR reduces the feature space from approximately 100-150 raw features to 30-40 optimized features[16], removing redundant timing measurements and movement characteristics that do not contribute to user discrimination.

C. Classification Approaches for Keystroke Dynamics

Traditional machine learning approaches have been widely applied to keystroke dynamics[11]. Support Vector Machines (SVMs) are effective in high-dimensional spaces and can model complex decision boundaries through kernel functions[6]. Random Forest classifiers provide robustness to noise and overfitting through ensemble averaging[14]. K-Nearest Neighbors (KNN) is simple and requires no training phase but can be computationally expensive at test time[20]. Deep learning approaches have gained substantial popularity in recent years[15]. Multilayer Perceptrons (MLPs) can model complex non-linear relationships but do not explicitly capture sequential structure[11]. Long Short-Term Memory (LSTM) networks are specifically designed to handle sequential data and can capture long-range dependencies in keystroke sequences[9]. Hybrid CNN+RNN architectures use convolutional layers to extract local features followed by recurrent layers to capture longer-range temporal dependencies[15].

Recent research has demonstrated that gradient boosting methods achieve state-of-the-art performance on keystroke authentication tasks[20]. A comparative study by Muralidharan and colleagues found that LightGBM achieved 94.68% accuracy on the CMU dataset[20], significantly outperforming prior hybrid approaches including the POHMM/SVM model which achieved 86.8% accuracy[20].

D. Dataset Collection for Behavioral Authentication Research

In addition to public benchmark datasets, controlled data collection is essential for evaluating authentication systems under specific application scenarios[7]. For

web-based authentication research, data collection is typically performed using a custom web application interface that simulates a banking or enterprise environment[18]. The data collection pipeline follows these steps:

- **User Interaction Recording:** Participants perform structured tasks (e.g., form filling, menu navigation, data entry) while keyboard and mouse events are captured using JavaScript event listeners[10]. Each event records timestamp, key code or mouse coordinates, and action type.
- **Sliding Window Segmentation:** The continuous event stream is divided into overlapping windows of fixed duration (e.g., 60 seconds) with 50% overlap between consecutive windows[14]. Each window generates a feature vector containing statistical summaries of keystroke timing (mean hold time, flight time variance) and mouse movement (average velocity, acceleration, click frequency)[11].
- **Multiple Sessions per User:** To capture natural behavioral variability, data is collected across multiple sessions on different days[13]. Each user contributes 3-5 sessions, with each session lasting 10-15 minutes of active interaction.
- **Impostor Data Collection:** To evaluate authentication performance, cross-user validation is performed where each user acts as an impostor for other users[10]. The dataset is balanced using SMOTE (Synthetic Minority Over-sampling Technique) to address the inherent class imbalance where genuine samples outnumber impostor samples[14].
- **Feature Normalization:** Per-user z-score normalization is applied to account for baseline differences in typing speed and mouse movement characteristics across users[11].

IV. MOUSE DYNAMICS AND MULTIMODAL AUTHENTICATION

Mouse dynamics analyzes the patterns of mouse movement and clicking behavior, offering complementary advantages to keystroke dynamics particularly in contexts where typing is infrequent[1], [10]. This section examines mouse dynamics-based authentication and multimodal systems that combine multiple behavioral signals[8], [11].

A. Segmentation and Feature Extraction for Mouse Dynamics

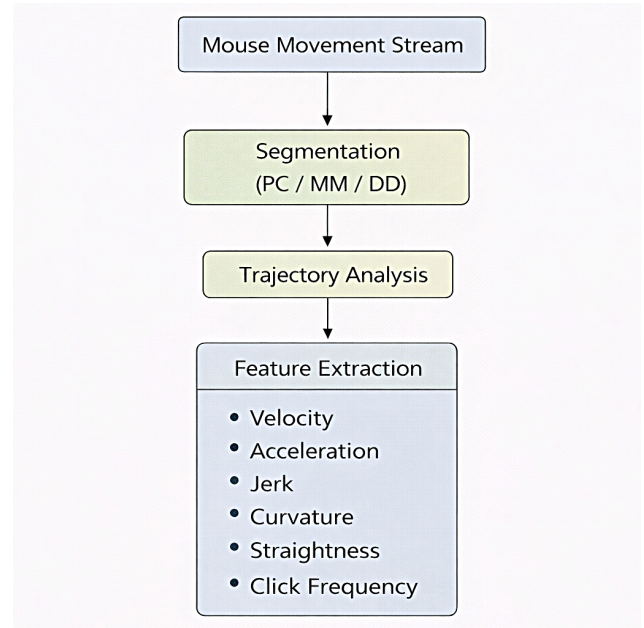


Fig. 5. Mouse movement segmentation and feature extraction pipeline for behavioral authentication.

A critical preprocessing step in mouse dynamics is segmentation, which divides the continuous stream of mouse events into meaningful logical units for feature extraction[1]. Point-and-Click (PC) segmentation identifies sequences of continuous mouse movement events that end with a click event[10]. Mouse-Move (MM) segmentation uses sequences of mouse movement events as segments without requiring a terminating click[1]. Drag-Drop (DD) segments contain sequences starting with a mouse down event, followed by multiple drag events, and concluding with a mouse release event[10].

Kinematic features capture the dynamics of mouse movement. Speed is calculated as the ratio of traveled distance to elapsed time[10]. Velocity captures the rate of position change and can be decomposed into horizontal and vertical components[1]. Acceleration captures the rate of change of velocity[10]. Jerk captures the rate of change of acceleration, providing information about movement smoothness[1]. Geometric features capture the shape and characteristics of movement trajectories. Straightness is the ratio between direct distance and actual trajectory length, with a value of 1 indicating perfectly straight movement[10]. Angle of curvature for any three consecutive points represents direction changes along the path[1].

B. Multimodal Systems Combining Keystroke and Mouse Dynamics

The fusion of keystroke and mouse dynamics into multimodal authentication systems offers substantial

advantages over unimodal approaches[8],[11]. When users interact with computers, they naturally switch between typing and mouse operations depending on the task at hand. A multimodal system can maintain continuous authentication during both typing-intensive tasks and mouse-intensive tasks, providing more consistent coverage[10]. Additionally, the two modalities capture different aspects of user behavior, and their combination can provide more robust and discriminative user models[11].

Research has demonstrated that combining keystroke and mouse dynamics yields better performance than either modality alone[10],[11]. One study found that performance on keystroke data alone was inadequate, while mouse behavior data achieved 85% accuracy through Random Forest, and combining both modalities increased performance to 90%[10]. The SIURUA method specifically addresses authentication in hybrid scenes where users perform diverse tasks by separating scene-related features from user-related features, achieving 84% accuracy in mixed-scenario environments[10].

V. MOBILE SENSOR-BASED AUTHENTICATION

The proliferation of smartphones and tablets equipped with diverse sensors has created new opportunities for behavioral biometric authentication[7],[8]. Modern mobile devices typically include accelerometers, gyroscopes, magnetometers, touchscreens, multiple cameras, GPS receivers, and various environmental sensors, enabling the capture of rich behavioral data[7].

A. Touchscreen Gesture Authentication

Touchscreen gestures represent the mobile analog of mouse dynamics, analyzing finger interaction patterns with capacitive touchscreens[7],[8]. Unlike mouse interactions, touchscreen interactions are direct and can capture pressure, contact area, and multi-touch gestures[5]. Common gestures analyzed include swipes, taps, flicks, pinches, and spreads[7]. Features extracted include velocity, acceleration, path deviation, pressure variation, touch area variation, and gesture duration[11].

B. Motion Sensor Authentication

Motion sensors, particularly accelerometers and gyroscopes, capture device movement and orientation during mobile device use[7]. The accelerometer measures proper acceleration in three spatial dimensions, capturing how users hold and move their devices[8]. The gyroscope measures angular velocity, capturing rotation and orientation changes[7]. Motion sensor-based authentication has been explored in various contexts

including picking up the phone, shaking gestures, in-air handwriting, and free-form gestures[5], [8].

C. Deep Learning for Mobile Authentication

Deep learning approaches have achieved state-of-the-art performance on mobile authentication tasks[11], [15]. A comprehensive evaluation comparing MLP, LSTM, bidirectional LSTM, and convolutional LSTM architectures on a mobile banking dataset found that deep learning models achieved exceptional performance on raw sensor and touch data, with accuracy exceeding 99.85% and FAR, FRR, and EER all below 0.5%[11]. Notably, performance on raw data substantially exceeded performance on handcrafted features, demonstrating the ability of deep networks to learn more discriminative representations directly from raw sensor streams[15].

VI. DEEP LEARNING APPROACHES

Deep learning has revolutionized behavioral biometric authentication by enabling automatic feature extraction from raw data[11], [15] and capturing complex temporal dependencies that traditional machine learning methods cannot represent[13].

A. Convolutional Neural Networks (CNN)

CNNs are particularly effective at extracting local patterns from sequential data[15]. In keystroke dynamics, CNNs can learn characteristic patterns from short sequences of key press events without requiring manual feature engineering[9]. In mouse dynamics, CNNs have been applied to image-based representations of mouse movement trajectories[10], treating the spatial path as an image to be classified[11].

B. Recurrent Neural Networks (RNN, LSTM, GRU)

Recurrent neural networks are specifically designed to handle sequential data[9], [11] and can capture dependencies across time steps. Long Short-Term Memory (LSTM) networks address the vanishing gradient problem that affects standard RNNs[15], enabling them to learn patterns over extended sequences[13]. Gated Recurrent Units (GRUs) provide a computationally efficient alternative to LSTMs with comparable performance[15]. In keystroke authentication, LSTMs can model the temporal evolution of typing patterns across entire sessions[9], [11].

C. Transformer Architectures

Transformer architectures, originally developed for natural language processing[13], have recently been applied to keystroke dynamics and have shown promising

results[13]. Transformers use self-attention mechanisms to capture dependencies between all positions in a sequence simultaneously[13], potentially identifying patterns that sequential RNNs might miss. Recent work has demonstrated that transformer-based models achieve extremely low EER below 0.02% on large-scale keystroke datasets[13].

D. Hybrid Architectures

Hybrid architectures combining multiple types of neural network layers have demonstrated state-of-the-art performance[15], [20]. CNN+RNN architectures use convolutional layers to extract local features followed by recurrent layers to capture longer-range temporal dependencies[15]. CNN+Transformer architectures combine local feature extraction with global attention mechanisms[13].

VII. PRIVACY-PRESERVING AUTHENTICATION

As behavioral biometric authentication systems collect increasingly rich data about user behavior[8],[13], privacy concerns become paramount[12]. Behavioral data can potentially reveal sensitive information including emotional state, health conditions, and demographic characteristics[7], [12].

A. Homomorphic Encryption for Behavioral Data

Homomorphic encryption enables computation on encrypted data without decrypting it[12]. In the context of behavioral authentication, a user can encrypt their behavioral feature vectors before sending them to an authentication server[12]. The server computes similarity scores (such as cosine similarity) on the encrypted data[12] and returns encrypted results, never seeing the raw biometric data[12]. The user then decrypts the results locally to determine authentication outcome. This approach ensures that even if the authentication server is compromised, raw behavioral data remains protected[12].

B. Secure Biometric Template Storage

Biometric templates stored on servers represent a significant security risk if compromised[8], [13]. Unlike passwords, compromised biometric data cannot be changed[8]. Secure template storage techniques including fuzzy commitment schemes and secure sketches enable template storage without exposing raw biometric data[12]. These techniques store only transformed or hashed versions of templates, and authentication is performed in the transformed domain[12].

C. Oblivious Transfer Protocols

Oblivious transfer protocols allow a user to retrieve specific information from a server without revealing which information was retrieved[12]. In behavioral authentication, oblivious transfer can be used to retrieve reference templates for only the specific activities a user performed[12], without revealing to the server which activities those were. This protects user activity privacy even from the authentication server[12].

VIII. BEHAVIORAL DRIFT AND ADAPTIVE AUTHENTICATION

Behavioral drift represents one of the most significant challenges facing behavioral biometric authentication systems[13], [21]. As user behavior evolves over time due to skill acquisition, fatigue, emotional state changes, aging, and device switching[13], static models trained on historical data become progressively less representative[16], leading to increased false rejection rates and degraded user experience[8].

A. Factors Contributing to Behavioral Drift

Skill acquisition is perhaps the most significant factor[13] as users become more experienced with typing, mouse use, or touchscreen interaction, they become faster, more consistent, and develop more efficient patterns. A user's typing speed may increase substantially over months or years of regular computer use[6]. Fatigue and circadian rhythms produce shorter-term fluctuations[21]: users type differently at different times of day, when tired versus alert, and when stressed versus relaxed. Emotional state also influences behavior[18], with studies demonstrating systematic changes in typing patterns under different emotional conditions. Illness and aging produce longer-term changes: neurological conditions affect motor control and typing patterns[13], and general aging slows reaction times. Device and environmental changes cause abrupt shifts[1], [10]: switching from a mechanical desktop keyboard to a laptop keyboard changes key travel distance and tactile feedback[1], affecting hold times and typing rhythm. Similarly, switching from an external mouse to a touchpad alters movement acceleration and trajectory patterns. An adaptive authentication system must implement device-aware modeling that detects such hardware changes[10] and adjusts behavioral profiles accordingly. Additionally, contextual factors including time of day, application type, and user posture (sitting versus standing) significantly influence behavioral patterns[21]. Systems that incorporate time-of-day context adaptation can maintain authentication accuracy by recognizing that a user typing at 9 AM after waking may exhibit different patterns than the same user typing at 9 PM after a full day of work[16].

B. Evidence of Limited Adaptability in Current Systems

A comprehensive survey of keystroke and mouse behavioral biometric systems found that only a small minority of studies (approximately 5-10%) implement any form of adaptation mechanism[13]. The vast majority evaluate authentication performance on data collected within a single session or across sessions separated by days or weeks, without assessing performance over longer periods during which behavioral drift would be expected to occur[2], [8]. This lack of adaptability has important practical consequences: a system that performs well on data collected within a single day may perform poorly when authenticating the same user months later[13], [21].

C. Existing Adaptation Mechanisms

Sliding window techniques maintain a fixed-size window of recent samples[16], discarding older samples as new samples become available, ensuring the user model reflects recent behavior. Growing window techniques accumulate all samples over time[13], retaining all historical information but potentially being slow to adapt to genuine behavioral changes. Transfer learning approaches train models on source data and then fine-tune them on new data from the target session[15], requiring fewer samples to adapt to new conditions. Double serial adaptation strategies use two thresholds to determine when to update user templates[10], adding clearly genuine samples to the reference set.

IX. PROPOSED HYBRID AUTHENTICATION ARCHITECTURE

Based on the comprehensive analysis of existing systems and identified research gaps[13], [8], we propose a Hybrid Adaptive Continuous Authentication (HACA) architecture designed specifically to address the challenges of behavioral drift, contextual variability, and long-term usability[13], [21].

A. Architecture Overview

The proposed architecture consists of three main layers operating in sequence: (1) a sliding window segmentation and multimodal feature extraction layer[16],[10] that processes raw input streams; (2) a hybrid classifier combining LightGBM for rapid classification with One-Class SVM for anomaly detection[20], [6]; and (3) an adaptive profile updating mechanism that incorporates incremental learning to track behavioral evolution[16], [13].

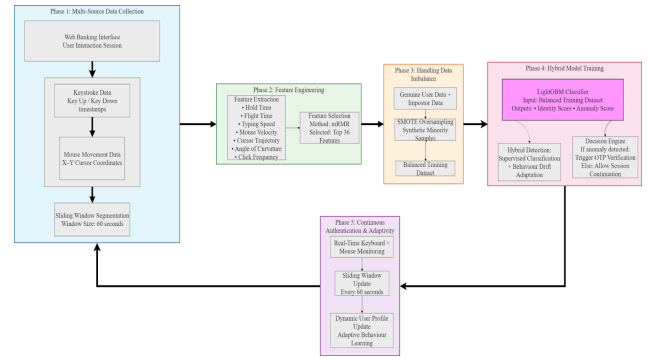


Fig. 6. Proposed hybrid adaptive continuous authentication architecture based on multimodal behavioral biometrics.

B. Layer 1: Sliding Window Segmentation and Feature Extraction

Raw input streams from keyboard, mouse, and (when available) touchscreen and motion sensors are divided into overlapping windows[16]. For keystroke data, each window is processed to extract hold times, flight times, digraph latencies[6], [13], typing speed, and error rates. For mouse data, each window yields movement velocity, acceleration, jerk, curvature, straightness[10], clicking speed, and clicking strength. Contextual features including device type, application in use, time of day[7], [21], and session duration are also extracted. All features are normalized per user[11] to account for baseline differences.

C. Layer 2: Hybrid Classifier with Anomaly Detection

The hybrid classifier employs two complementary mechanisms operating in parallel[14] with distinct roles:

LightGBM serves as the primary supervised classifier[20] for rapid identity verification. Trained on labeled genuine user samples, LightGBM outputs a probability score indicating whether the current behavioral window belongs to the legitimate user[20]. With demonstrated accuracy of 94.68% on keystroke benchmark datasets[20], LightGBM provides fast, reliable classification for routine authentication decisions.

One-Class SVM serves as an anomaly detection boundary validator [6], [18] that operates without requiring impostor training samples[6]. Trained exclusively on legitimate user behavior, the One-Class SVM learns the boundary of normal behavioral patterns[18]. When current behavior falls outside this learned boundary, it signals potential behavioral drift or impostor activity. This is particularly valuable for detecting novel attacks or behavioral patterns not represented in training data[18].

Decision Fusion: The final authentication decision integrates both outputs using a confidence-weighted mechanism[14]:

- If LightGBM confidence is high (>0.85) and One-Class SVM confirms normalcy, authentication is granted
- If LightGBM confidence is moderate but One-Class SVM indicates anomaly, additional verification is triggered
- If LightGBM confidence is low regardless of anomaly status, authentication is rejected

This dual-classifier approach balances security (anomaly detection catches what supervised learning misses) with usability (LightGBM provides fast, accurate decisions for routine cases)[8].

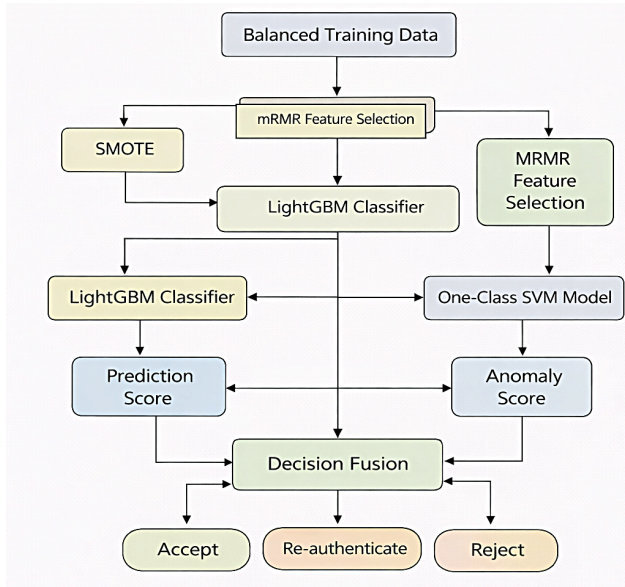


Fig. 7. Hybrid classification framework combining LightGBM prediction with One-Class SVM anomaly detection.

D. Layer 3: Adaptive Profile Updating

When the system detects behavioral drift through sustained anomalies[13] or through verification of genuine user activity following false rejections, the adaptive updating mechanism is triggered. The system collects recent behavioral samples as new training data and performs incremental learning to update the user profile[16]. The update process is carefully controlled to adapt to genuine behavioral evolution while remaining robust to temporary fluctuations[13]. The update rate is adaptive: during periods of stable behavior, updates are minimal; during periods of rapid behavioral change (such

as after device switching)[21], updates are more aggressive.

E. Advantages of the Proposed Architecture

The proposed architecture offers several advantages over existing approaches. First, it explicitly addresses behavioral drift [13] through the combination of short-term classification and long-term anomaly detection with adaptive updating. Second, the hybrid classifier provides both rapid decision-making and consistency monitoring[14], balancing security and usability. Third, the sliding window approach enables real-time operation[16] with minimal latency. Fourth, the architecture is modality-agnostic[8] and can incorporate additional behavioral signals as they become available. Fifth, the adaptive updating mechanism enables the system to evolve with the user over time[13], [16] without requiring complete retraining.

X. CONCLUSION

Behavioral biometric authentication provides a scalable, secure, and user-friendly alternative to traditional authentication methods[8], [13], enabling continuous identity verification throughout active sessions without interrupting user workflow[7], [11]. This survey has provided a comprehensive analysis of keystroke dynamics, mouse dynamics, touchscreen gestures, and sensor-based behavioral signals, along with the machine learning and deep learning techniques employed for identity verification[11], [15].

Key findings from this survey include: (1) deep learning approaches, particularly LSTM and hybrid CNN+RNN architectures, achieve state-of-the-art performance on behavioral authentication tasks[15], with accuracy exceeding 99% on mobile sensor data[11]; (2) gradient boosting methods such as LightGBM achieve exceptional performance on keystroke dynamics[20], reaching 94.68% accuracy on benchmark datasets[20]; (3) multimodal fusion combining keystroke and mouse dynamics improves robustness and accuracy compared to unimodal systems[10], [11]; (4) behavioral drift due to skill acquisition, fatigue, device switching, and aging remains a significant challenge[13], [21], with only a small minority of current systems implementing adaptation mechanisms[13]; (5) privacy-preserving techniques including homomorphic encryption enable secure authentication without exposing raw behavioral data[12].

Future systems should integrate contextual awareness to account for task, device, and environmental

variations[21]; incorporate adaptive learning mechanisms to track behavioral evolution over extended periods[16]; adopt privacy-preserving architectures to protect sensitive behavioral data[12]; and leverage multimodal fusion to improve robustness across diverse usage scenarios[10]. The hybrid adaptive authentication architecture proposed in this survey provides a framework for addressing these requirements, combining real-time classification with anomaly detection and incremental profile updating to maintain authentication accuracy as user behavior naturally evolves over time[16], [20].

REFERENCES

- [1] Z. Jorgensen and T. Yu, "On mouse dynamics as a behavioral biometric for authentication," in *Proc. 6th ACM Symp. Information, Computer and Communications Security (ASIACCS)*, 2011, pp. 476–482.
- [2] P. S. Teh, A. B. J. Teoh, and S. Yue, "A survey of keystroke dynamics biometrics," *Scientific World Journal*, 2013.
- [3] Y. Zhong and Y. Deng, "A survey on keystroke dynamics biometrics: approaches, advances, and evaluations," *Recent Advances in User Authentication Using Keystroke Dynamics Biometrics*, 2015.
- [4] B. Singh, S. Sonawane, Y. Shah, and V. Singh, "Literature survey on keystroke dynamics for user authentication," *International Journal on Recent and Innovation Trends in Computing and Communication*, vol. 5, no. 5, pp. 280–282, 2017.
- [5] M. Antal et al., "A survey on behavioral biometric authentication on smartphones," *Journal of Information Security and Applications*, vol. 37, pp. 28–37, 2017.
- [6] S. Krishnamoorthy, L. Rueda, S. Saad, and H. Elmiligi, "Identification of user behavioral biometrics for authentication using keystroke dynamics and machine learning," in *Proc. 2nd Int. Conf. Biometric Engineering and Applications (ICBEA)*, 2018, pp. 50–57.
- [7] M. Abuhamad, A. Abusnaina, D. Nyang, and D. Mohaisen, "Sensor-based continuous authentication of smartphones' users using behavioral biometrics: A contemporary survey," arXiv preprint arXiv:2001.08578, 2020.
- [8] A. A. Ahmad et al., "Behavioral biometrics & continuous user authentication on mobile devices: A survey," *Information Fusion*, vol. 66, pp. 76–99, 2021.
- [9] P. Kasproski, Z. Borowska, and K. Harezlak, "Biometric identification based on keystroke dynamics," *Sensors*, vol. 22, no. 9, p. 3158, 2022.
- [10] X. Wang, Y. Shi, K. Zheng, Y. Zhang, W. Hong, and S. Cao, "User authentication method based on keystroke dynamics and mouse dynamics with scene-irrelevant features in hybrid scenes," *Sensors*, vol. 22, no. 17, p. 6627, 2022.
- [11] U. Uslu, Ö. D. İncel, and G. I. Alptekin, "Evaluation of deep learning models for continuous authentication using behavioral biometrics," *Procedia Computer Science*, vol. 225, pp. 1272–1281, 2023.
- [12] A. F. Baig, S. Eskeland, and B. Yang, "Privacy-preserving continuous authentication using behavioral biometrics," *International Journal of Information Security*, vol. 22, pp. 1833–1847, 2023.
- [13] R. Shadman et al., "Keystroke dynamics: concepts, techniques, and applications," *ACM Computing Surveys*, 2023/2025.
- [14] X. Wang and D. Hou, "Enhancing keystroke dynamics authentication with ensemble learning and data resampling techniques," *Electronics*, vol. 13, no. 22, p. 4559, 2024.
- [15] Ł. Wycislik, P. Wyleżek, and A. Momot, "The improved biometric identification of keystroke dynamics based on deep learning approaches," *Sensors*, vol. 24, no. 12, p. 3763, 2024.
- [16] P. Bansal and A. Ouda, "Continuous authentication in the digital age: An analysis of reinforcement learning and behavioral biometrics," *Computers*, vol. 13, no. 4, p. 103, 2024.
- [17] X. Liu et al., "SoundAuth: Non-intrusive user authentication via keystroke acoustic biometrics with triplet networks," University of Electronic Science and Technology of China, 2024/2025.
- [18] A. Almeahmadi, "Cyber coercion detection using LLM-assisted multimodal biometric system," *Applied Sciences*, vol. 15, no. 19, p. 10658, 2025.
- [19] A. Martins et al., "Keystroke dynamics for intelligent biometric authentication with machine learning," *Discover Applied Sciences*, vol. 7, p. 992, 2025.
- [20] A. Muralidharan et al., "A comparative analysis of machine learning models for behavioral biometric authentication using keystroke dynamics," *Procedia Computer Science*, vol. 265, pp. 116–123, 2025.
- [21] K. Asgarov, "Endpoint behavior modelling from user interaction metrics via unsupervised deep learning for anomaly detection," Azerbaijan Technical University, 2025.
- [22] A. T. Vaidya and J. D. Mhatre, "Behavioural biometrics for continuous authentication in cybersecurity systems," *International Journal of Computer Technology and Electronics Communication*, 2025.

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